

《计算机网络原理》

课程编号：40240513

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本科生必修课

计算机科学与技术系

主要教学内容和学时分配

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第三章	数据通信的基本原理	3
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第六章 局域网和介质访问控制

主要内容

- 6.0 局域网技术概述
- 6.1 通道分配问题
- 6.2 多路访问协议
- 6.3 局域网和城域网标准
 - 以太网Ethernet: 802.3
- 6.4 无线局域网 : 802.11
- 6.5 宽带无线局域网: 802.16
- 6.6 桥接器、交换机和路由器

6.0 局域网技术概述

- 产生局域网的原因
 - 1980年代，微型机发展迅速，彼此需要相互通信（近距离），共享资源；
 - 分布式的网络应用：分布式计算，分布式数据库
- 定义
 - 局域网是一种将小区域内的各种通信设备互连在一起的通信网络。
- 有线和无线局域网
 - 有线：Ethernet
 - 无线：WiFi

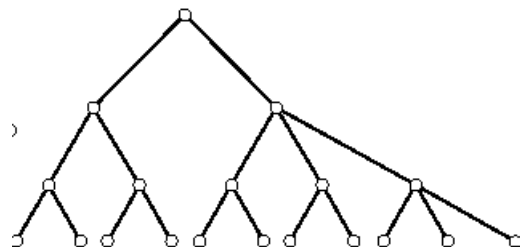
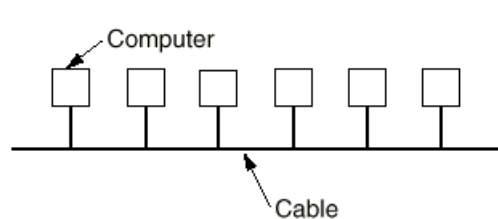
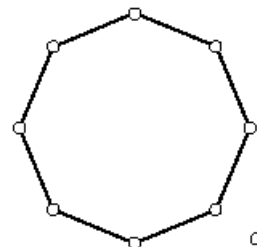
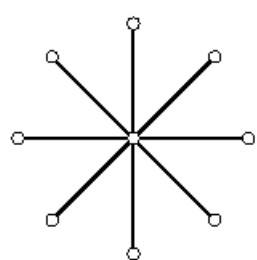
6.0 局域网技术概述

- 局域网的基本特点
 - 高速率 (10 Mbps ~ 100 Gbps)
 - 短距离 (0.1 km ~ 10 km)
 - 低错率 (10^{-8} ~ 10^{-11})
- 局域网发展趋势
 - 有线局域网: 400G Ethernet (IEEE 802.3bs)
 - 无线局域网: 2018年全球40亿台设备支持WiFi
 - WiFi 3 (IEEE 802.11g) , 54Mbps
 - WiFi 4 (IEEE 802.11n) , 600Mbps
 - WiFi 5 (IEEE 802.11ac) , 6.9Gbps
 - WiFi 6 (IEEE 802.11ax) , 9.6Gbps

6.0 局域网技术概述

■ 局域网拓扑结构

- 星型结构
- 环型结构
- 总线型结构
- 树型结构



■ 传输介质

- 双绞线
- 基带同轴电缆
- 光纤
- 无线

6.1 通道分配问题

- 局域网的关键技术问题
 - 一般使用广播通信。
 - 如何分配广播通道？
- 静态通道分配：LANs and MANs
 - FDM：频分多路复用
 - TDM：时分多路复用
- 动态通道分配：LANs and MANs
 - 通道分配模型及其假设

动态通道分配

1. Station Model 独立站点

- 假设：每个站点是独立的，并以统计固定的速率产生帧，一帧产生后到被发送走之前，站点被封锁

2. Single Channel Assumption 单一通道假设

- 所有的通信都是通过单一的通道来完成的，各个站点都可以从通道上收发信息

3. Collision Assumption. (冲突) 碰撞假设

- 若两帧同时发出，会相互重叠，结果使信号无法辨认，称为冲突。所有的站点都能检测到冲突，冲突帧必须重发

4. 何时发送：连续时间和分槽时间？

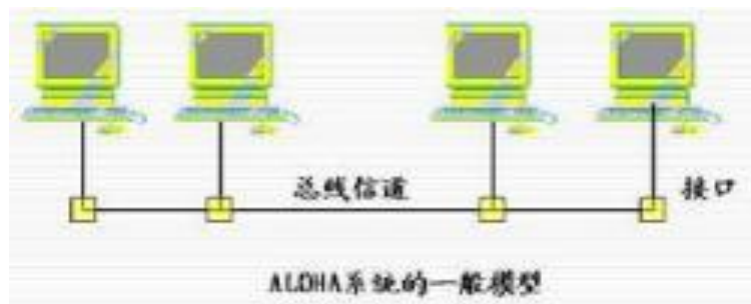
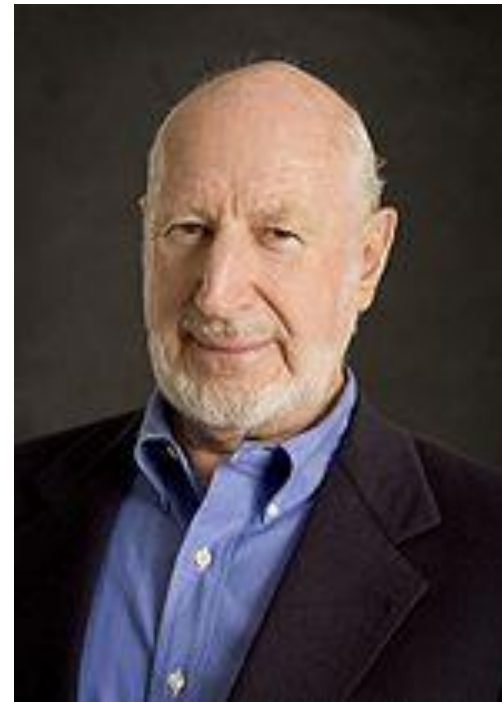
5. 能否发送：载波监听和非载波监听？

6.2 多路访问协议

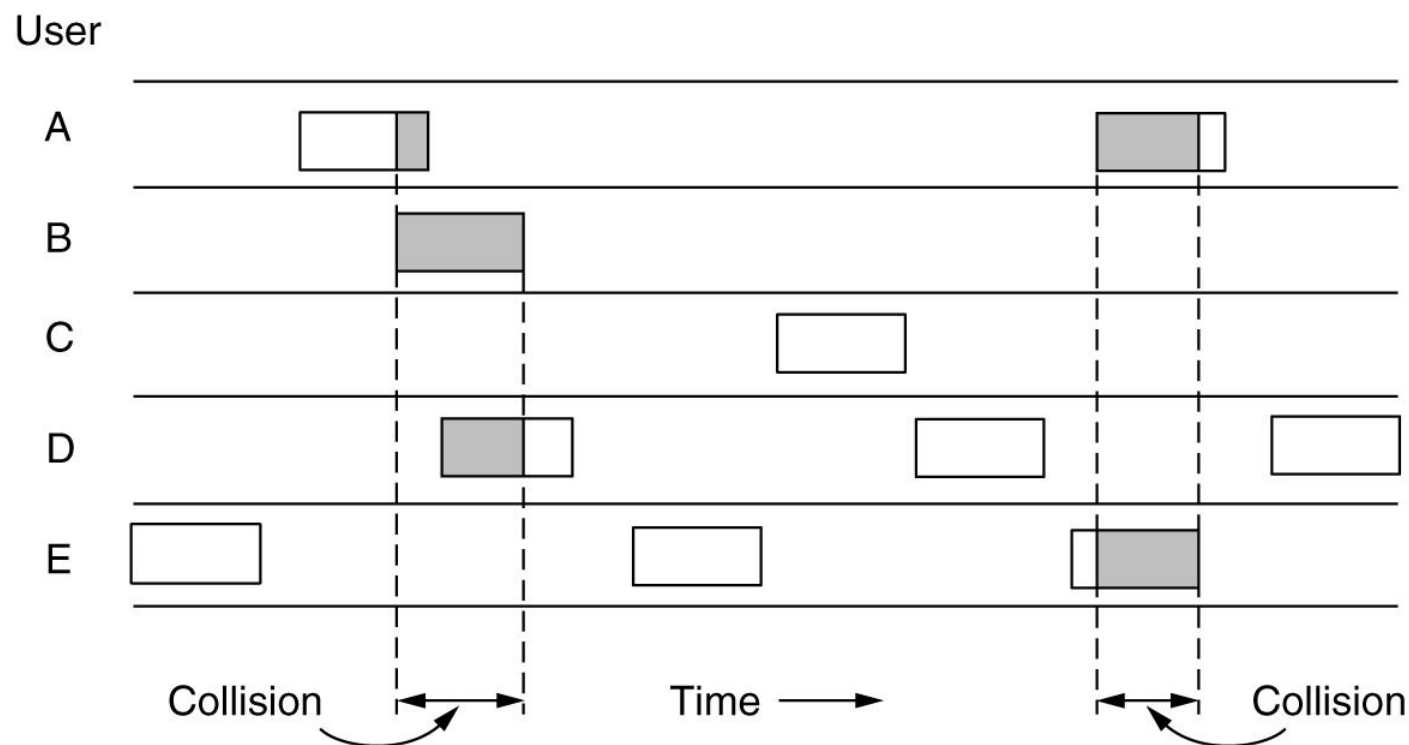
- ALOHA
- Carrier Sense Multiple Access Protocols (CSMA) 载波监听多路访问协议CSMA
- 带冲突检测的载波监听多路访问协议CSMA/CD
- Collision-Free Protocols 无冲突协议
- Limited-Contention Protocols有限竞争协议
- Wavelength Division Multiple Access Protocols 波分多路访问协议
- Wireless LAN Protocols无线局域网协议

ALOHA协议

- 1970年代，夏威夷大学的Norman Abramson设计了ALOHA协议
- 目的：解决信道的动态分配
 - 基本思想可用于任何无协调关系的用户争用单一共享信道使用权的系统
- 分类
 - 纯ALOHA协议
 - 分槽ALOHA协议



纯 ALOHA (1)

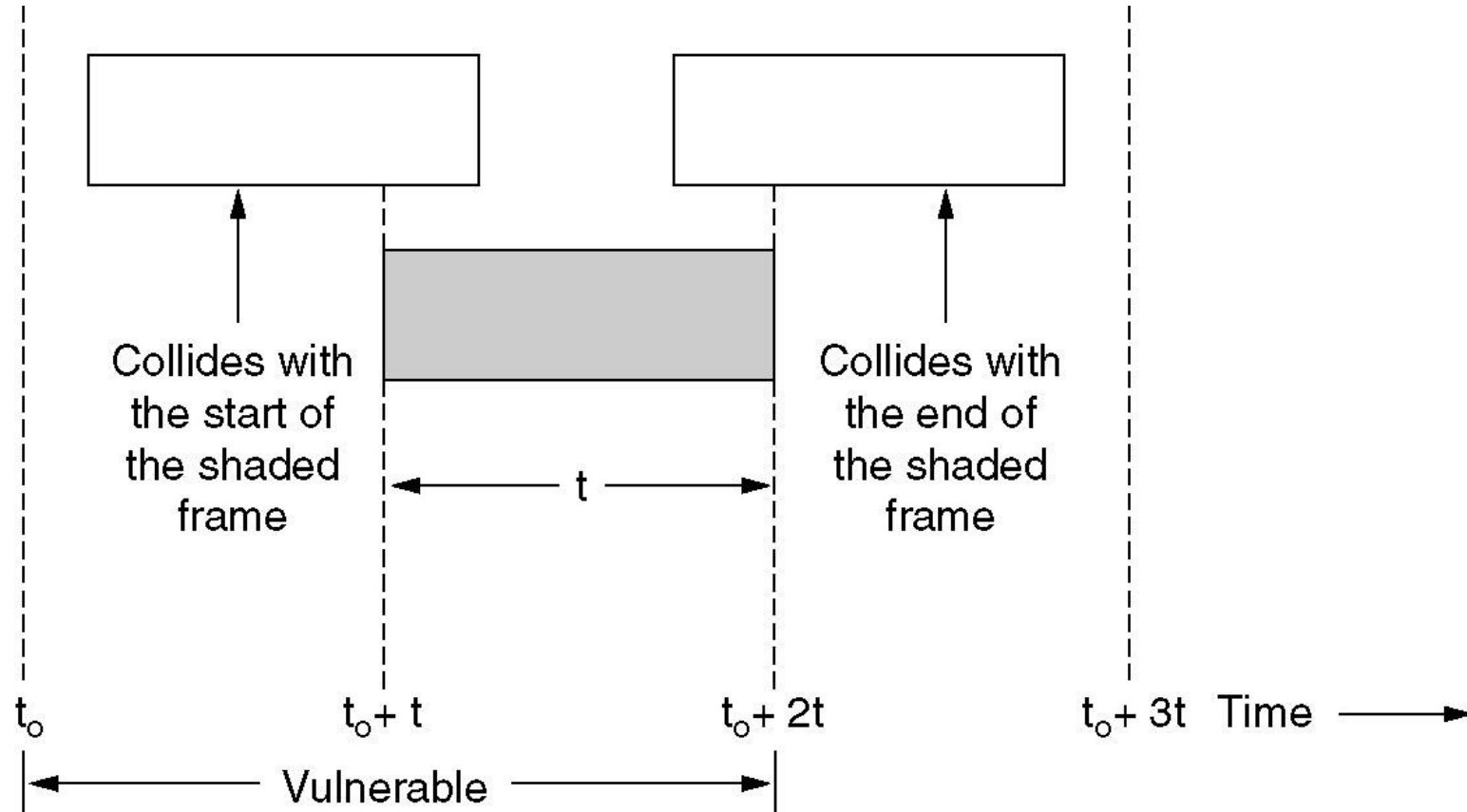


- 工作原理：站点只要产生帧，就立即发送到信道上；规定时间内若收到应答，表示发送成功，否则重发。
- 重发策略：等待一段随机的时间，然后重发；如再次冲突，则再等待一段随机的时间，直到重发成功为止
- 优点：简单易行
- 缺点：极容易冲突

纯 ALOHA (2)

- 基本思想：用户有数据要发送时，可以直接发至信道；然后监听信道看是否产生冲突，若产生冲突，则等待一段随机的时间重发
- 多用户共享单一信道，并由此产生冲突，这样的系统称为竞争系统
- 信道效率：
 - 假设：帧长固定，无限个用户，按泊松分布产生新帧，平均每个帧时 (frame time) 产生 S 帧 ($0 < S < 1$)；发生冲突重传，新旧帧共传 k 次，遵从泊松分布，平均每个帧时 (时间槽) 产生 G 帧
 - 吞吐率 $S = GP_0$ ， P_0 为发送一帧不受冲突影响的概率
 - 冲突危险区
 - 一个帧时内产生 k 帧的概率服从泊松分布： $\Pr[k] = \frac{G^k e^{-G}}{k!}$ ，两个帧时平均产生 $2G$ 个帧，在冲突危险区内无其它帧产生的概率为： $P_0 = e^{-2G}$ ，所以 $S = Ge^{-2G}$
 - 效率：最高信道利用率 (极大值) 为 $G = 0.5$ 时， $S = 0.18418.4\%$

纯 ALOHA (3)

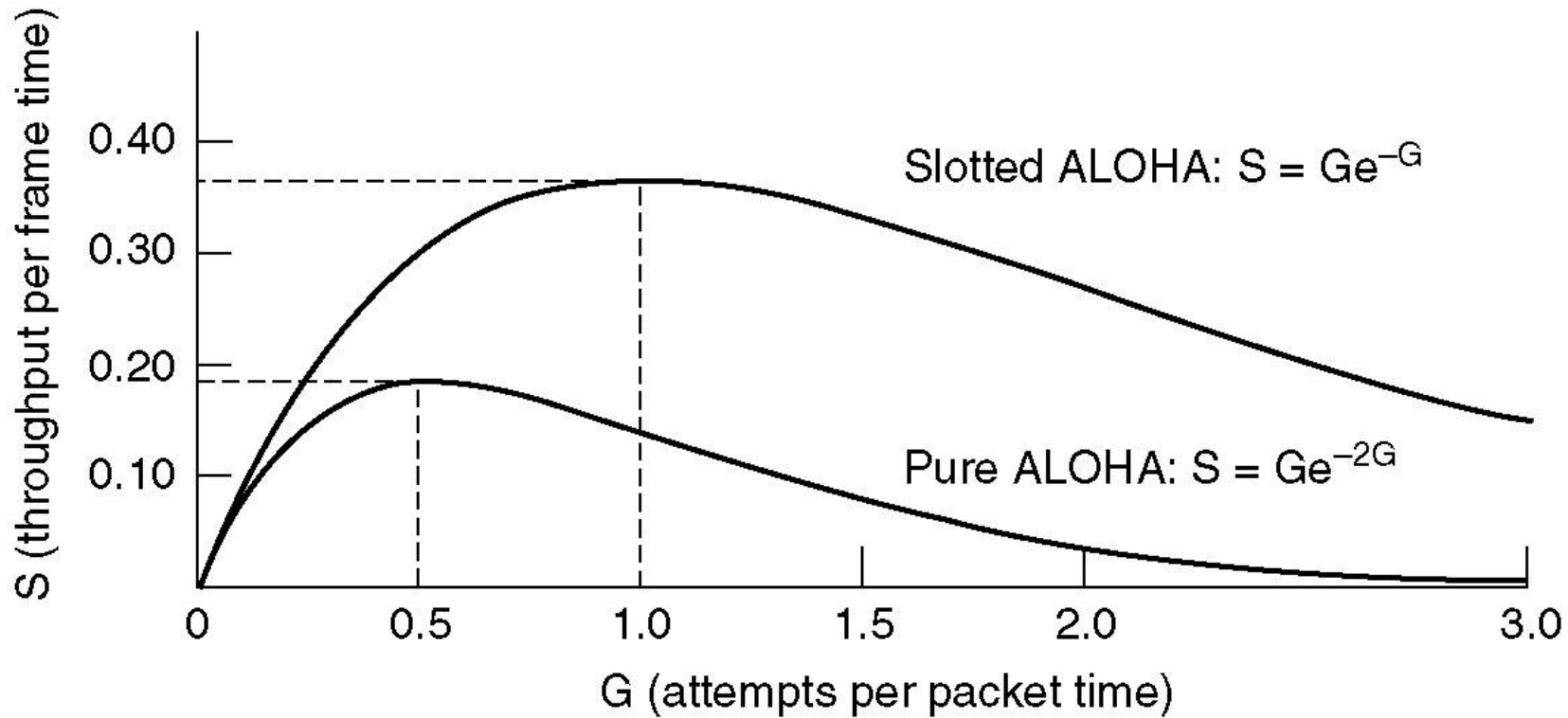


Vulnerable period for the shaded frame.

分槽ALOHA

- 基本思想：把信道时间分成离散的时间槽，槽长为一个帧所需的发送时间。每个站点只能在时槽开始时才允许发送。其他过程与纯ALOHA协议相同
- 重发策略：
 - 与纯ALOHA相同，等待一段随机的时间，然后重发；如再次冲突，则再等待一段随机的时间，直到重发成功为止。
- 信道效率
 - 冲突危险区是纯ALOHA的一半，所以 $P_0 = e^{-G}$ ， $S = Ge^{-G}$
 - 与纯ALOHA协议相比，分槽ALOHA降低了产生冲突的概率，信道利用率最高为36.8%

ALOHA系统吞吐量和负载的关系



Throughput versus offered traffic for ALOHA systems.

载波监听多路访问CSMA

- CSMA 基本原理

- 站点在发送数据之前先监听信道。如果信道空闲，站点就可以发送数据；如果信道忙，则站点不能发送数据。
- 如果两个站点都检测到信道是空闲的，并且同时开始传送数据，那么这几乎会立即导致冲突。
- 站点在监听信道时，听到信道是空闲的，但这并不意味着信道真的空闲，因为其他站点的数据此时可能正在信道上传送，但由于传播时延，信号还没有到达正在监听的站点，从而引起对信道状态的错误判断。
- 信道空闲时，是否立即发送？ X 坚持型CSMA

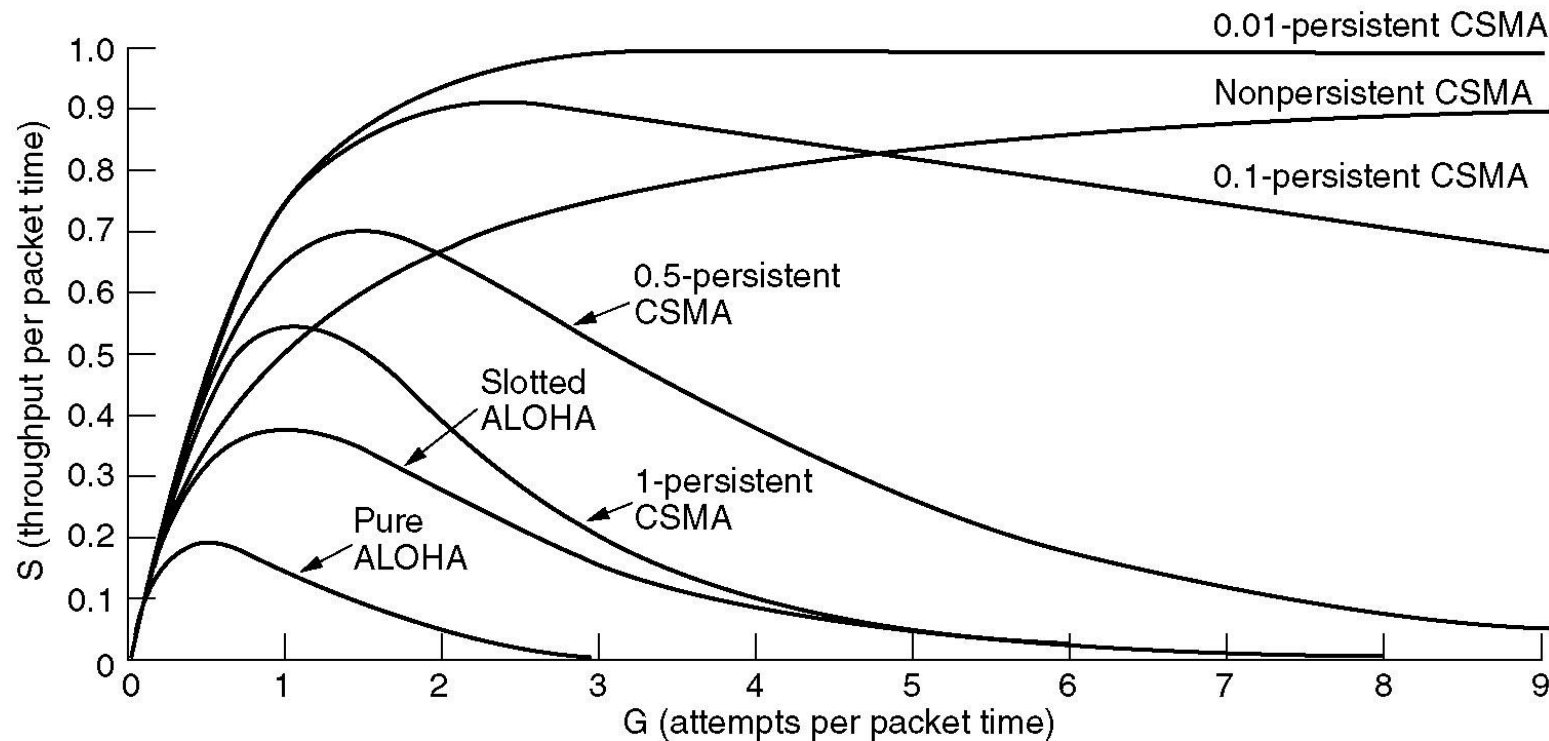
- 带冲突检测的载波监听多路访问 CSMA/CD

- 发送站点在发送过程中仍继续侦听介质，以检测是否存在冲突。

载波监听多路访问CSMA

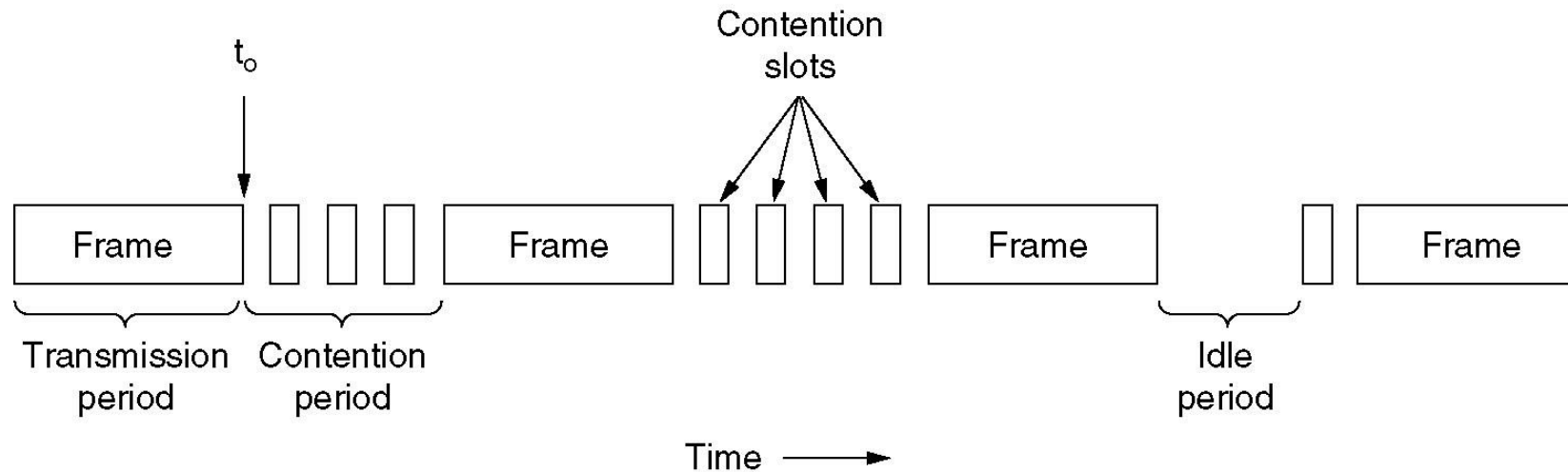
- CSMA 基本原理
- 1 - 坚持型 CSMA
 - 发送前监听，通道空，发送；通道不空，继续监听，直到通道空，发送。
优：通道利用率高；缺：冲突大。
- 非坚持型 CSMA
 - 发送前监听，通道空，发送；通道不空，不继续监听，等待一个概率时间，继续监听，直到通道空，发送。
 - 优：通道利用率高，冲突小；缺：延迟大。
- P-坚持型 CSMA（分时间槽通道）
 - 发送前在时间片槽前监听，通道空，以概率 p 发送；通道不空，继续监听，等待下一个时间片槽，继续监听。

坚持型和非坚持型的CSMA



Comparison of the channel utilization versus load for various random access protocols.

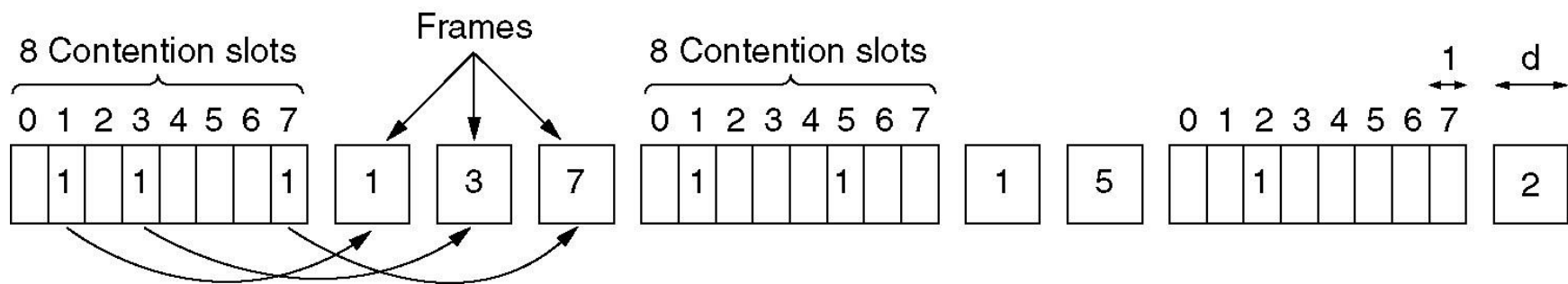
帶冲突检测的载波监听多路访问CSMA/CD (CSMA with Collision Detection)



CSMA/CD can be in one of three states: contention, transmission, or idle.

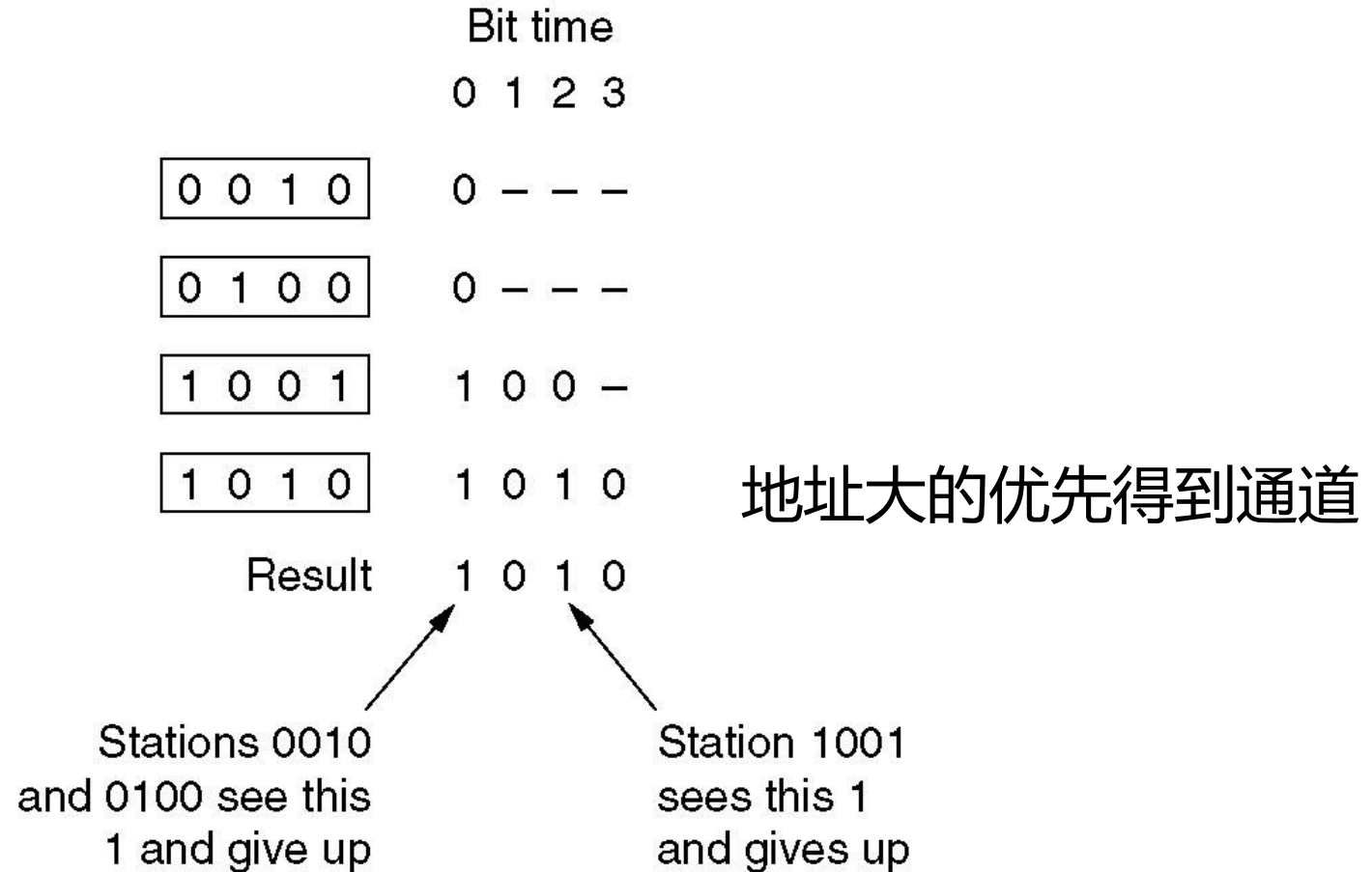
无冲突协议

Collision-Free Protocols



基本位图协议：The basic bit-map protocol

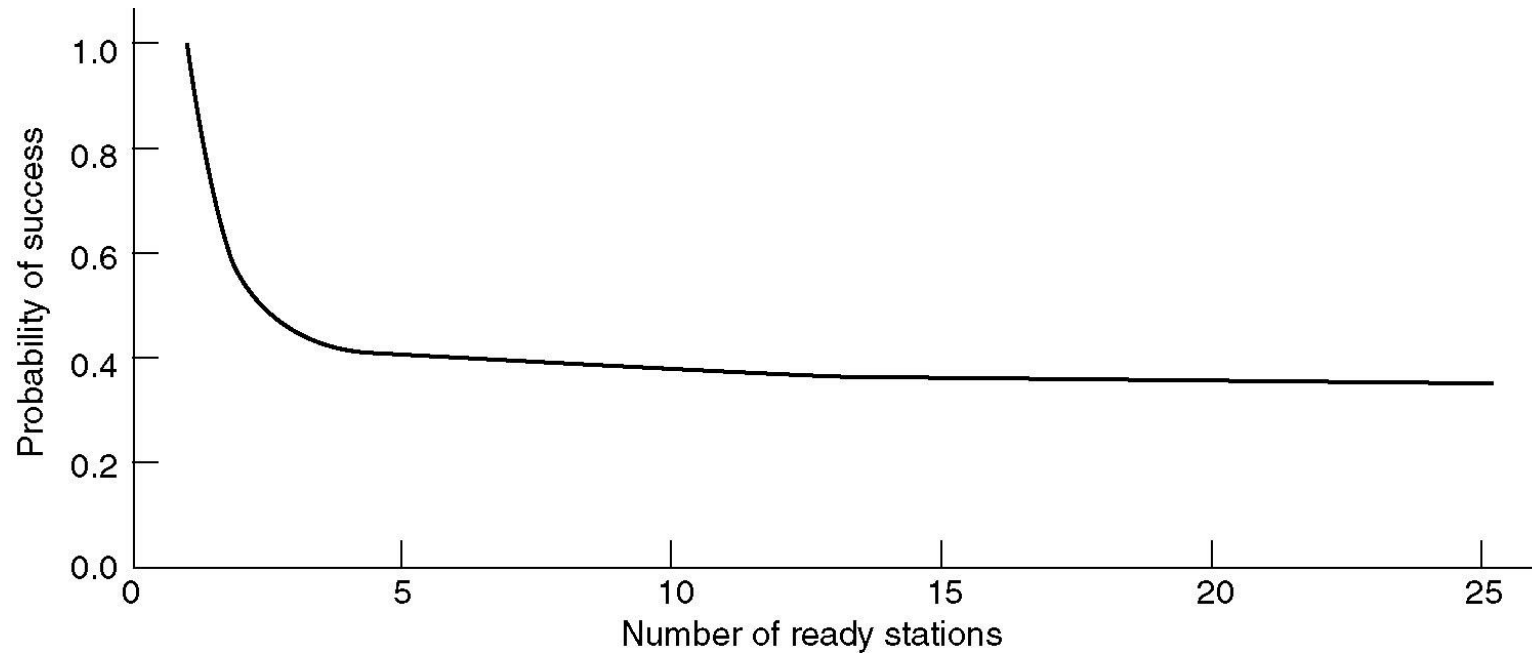
无冲突协议 (Collision-Free Protocols)



二进制倒数协议: The binary countdown protocol
A dash indicates silence

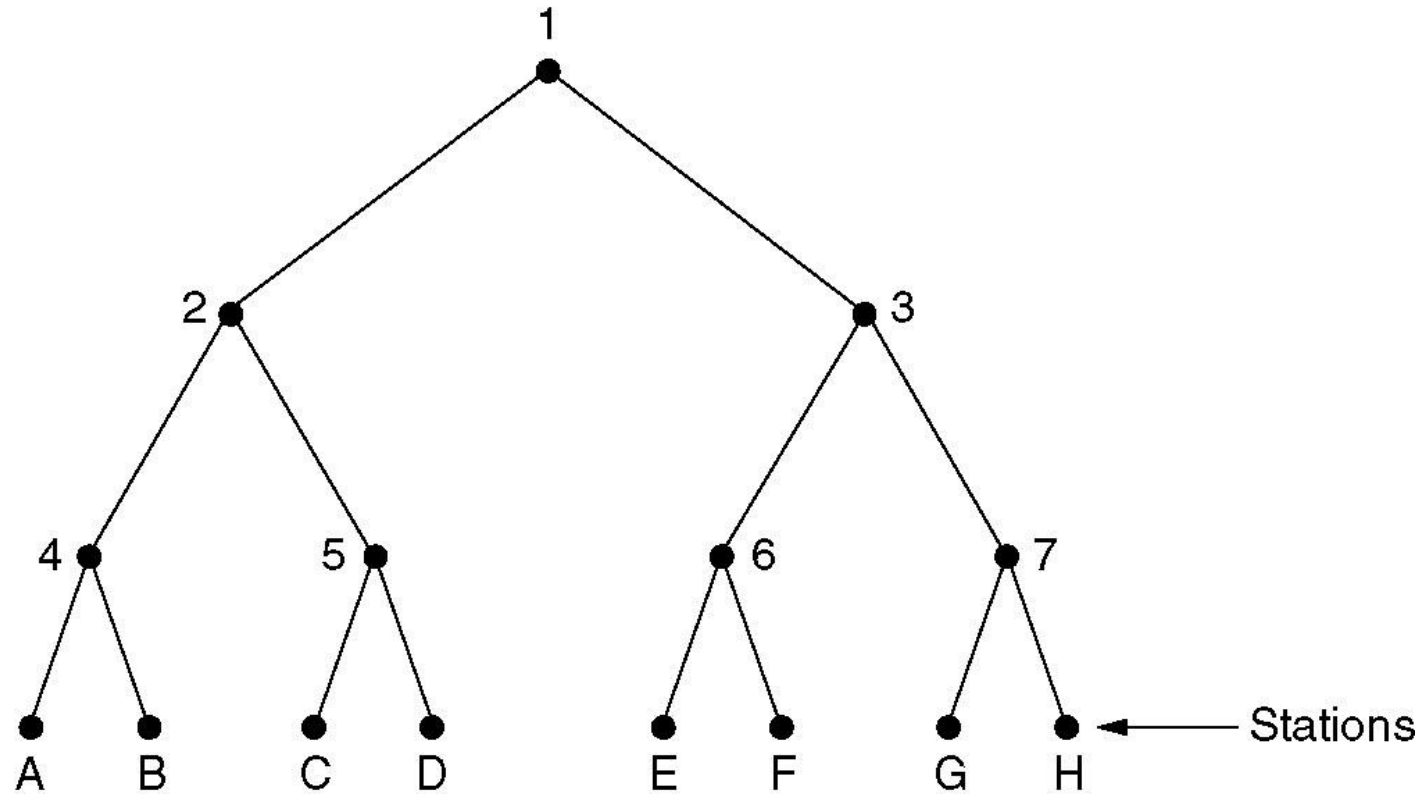
有限争抢协议

Limited-Contention Protocols



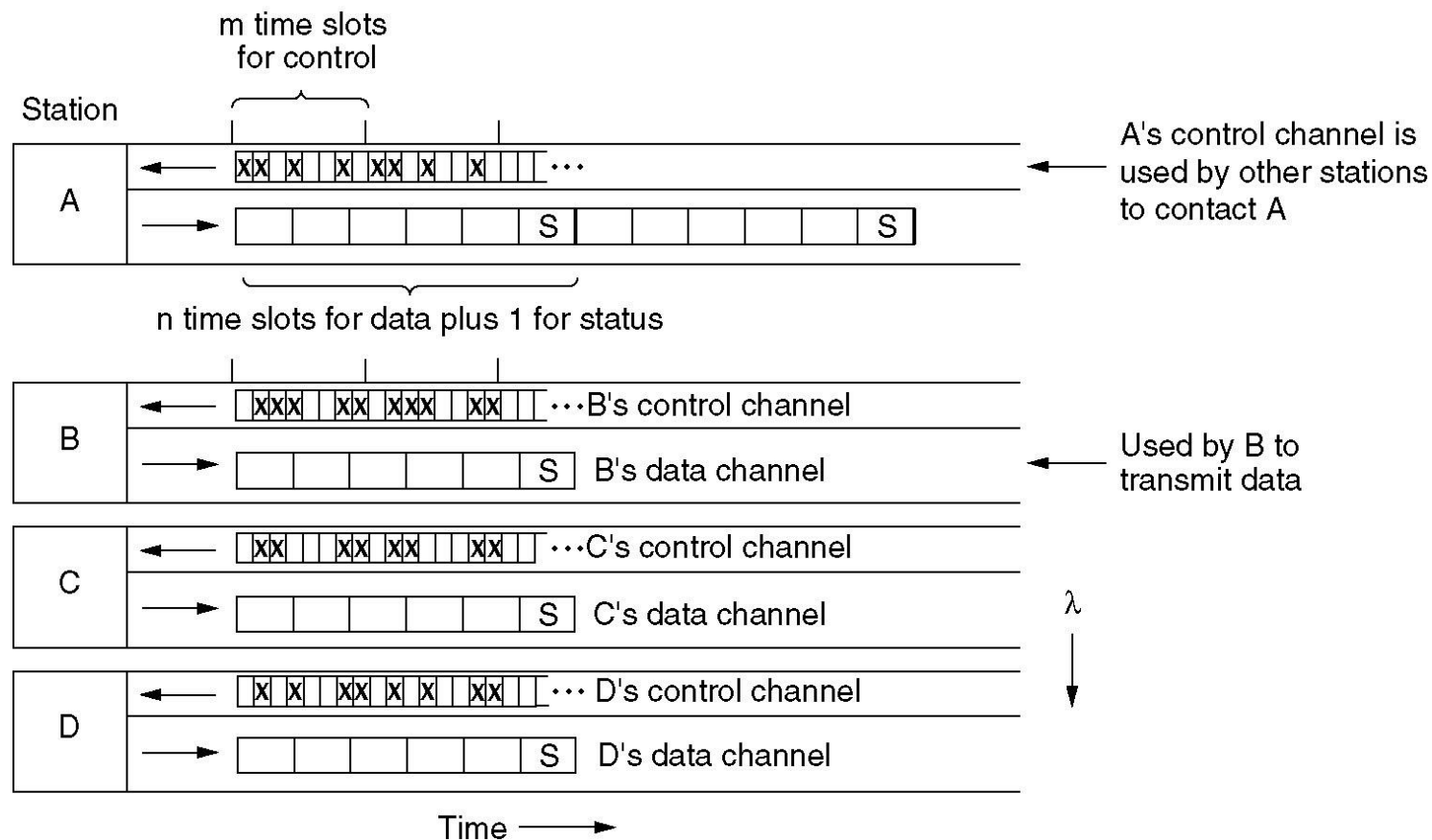
结合竞争协议和无冲突协议：对称竞争信道的获得概率

自适应树遍历协议 Adaptive Tree Walk Protocol



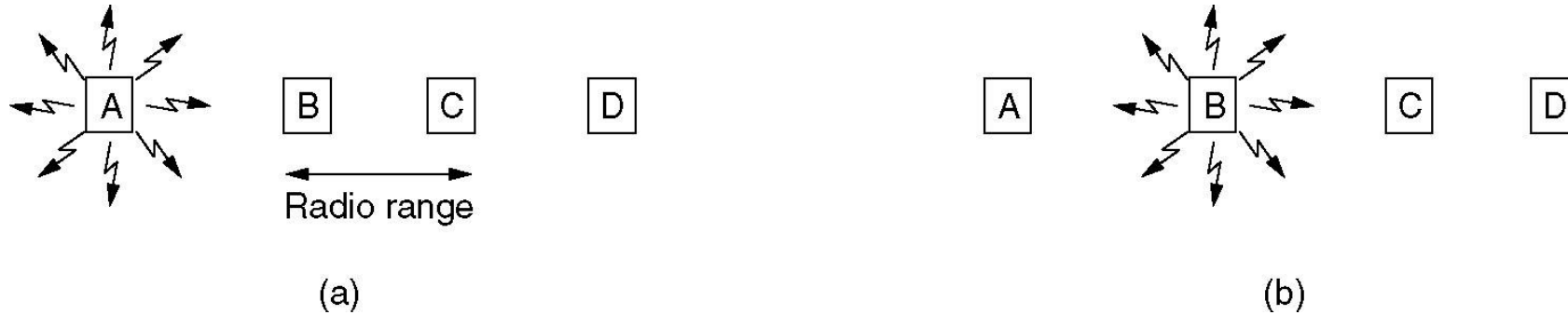
自适应树遍历协议

波分多路访问协议



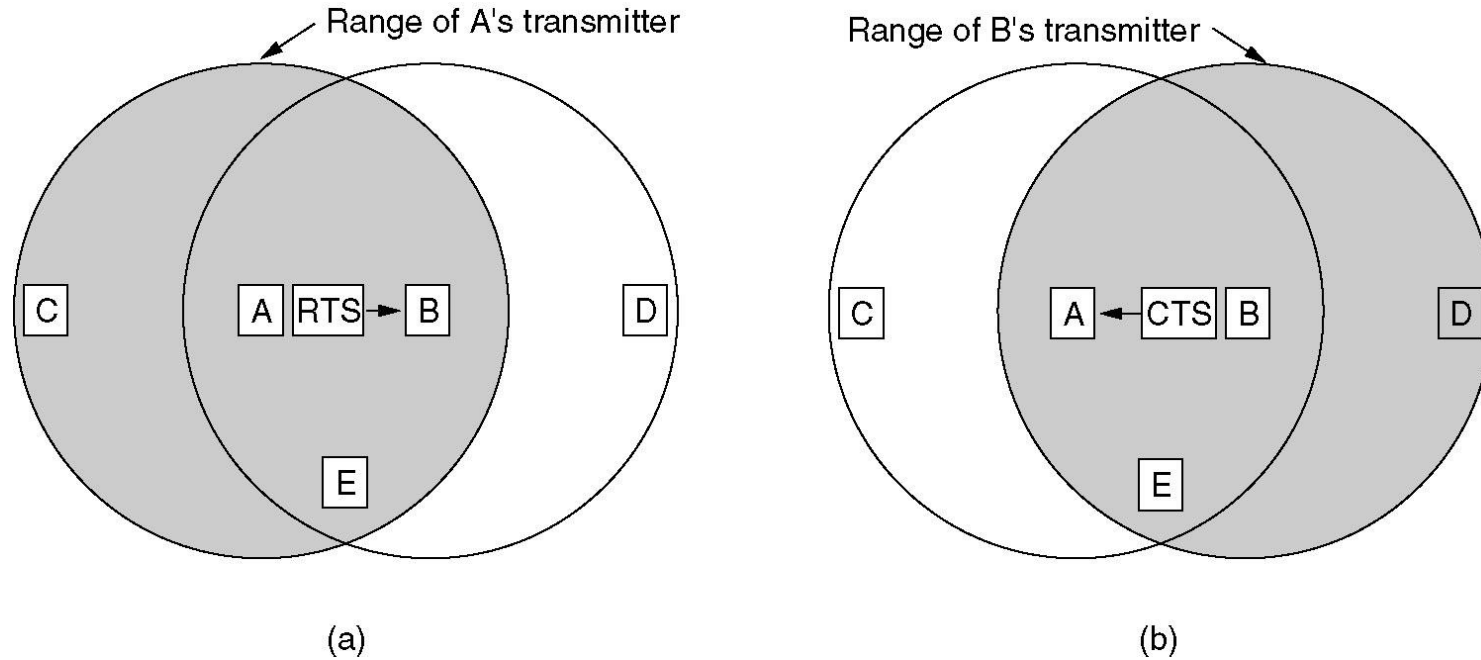
Wavelength division multiple access.

无线局域网 (1)



A wireless LAN. (a) A transmitting. (b) B transmitting.

无线局域网 (2)



The MACA protocol.避免冲突的多路访问协议

(a) A sending an RTS to B. (b) B responding with a CTS to A.

6.3 局域网和城域网标准

- IEEE 802.2 LLC 协议
- IEEE 802.3 和 Ethernet
- IEEE 802.4 和 Token Bus
- IEEE 802.5 和 Token Ring
- IEEE 802.11 无线局域网协议
- IEEE 802.16 宽带无线城域网

以太网 (Ethernet)

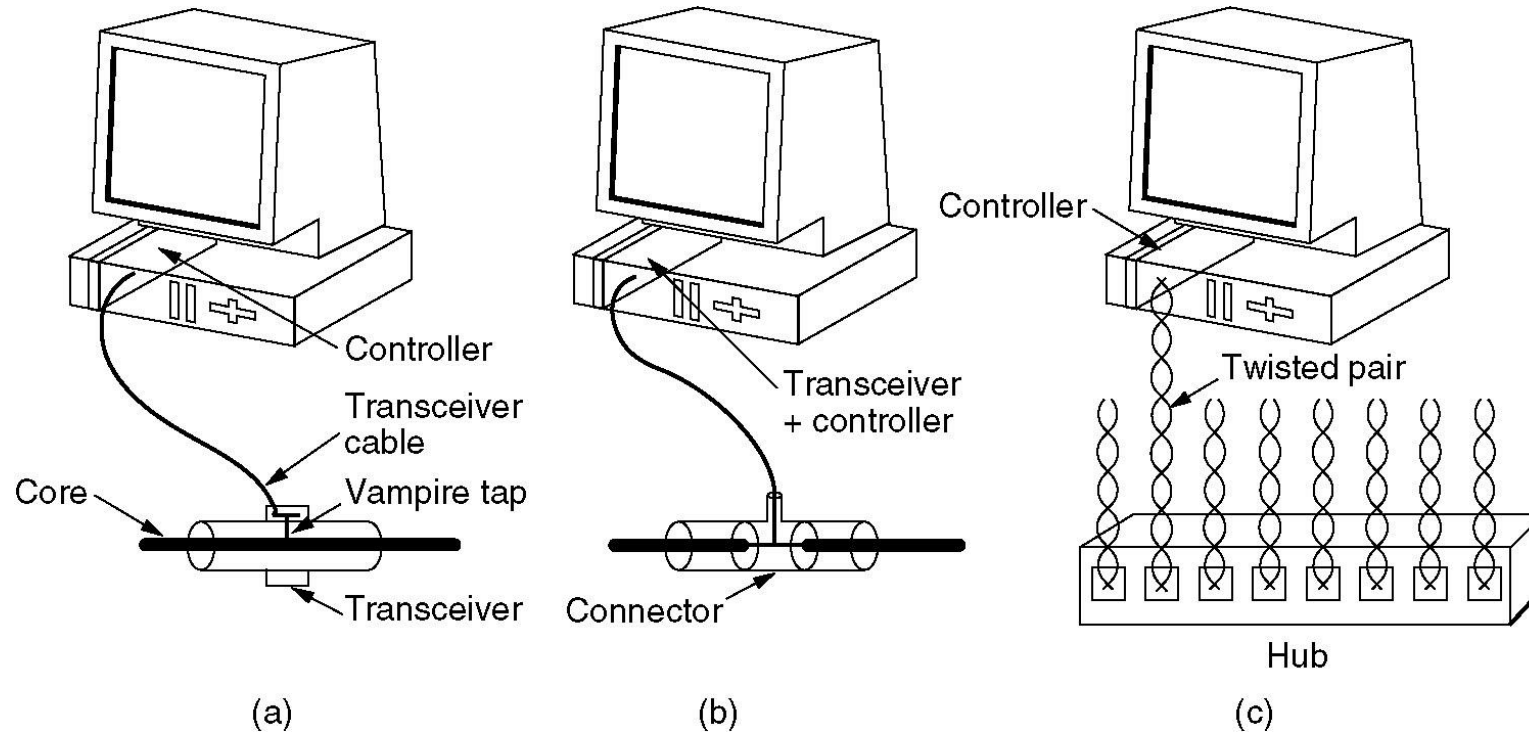
- Ethernet Cabling
- Manchester Encoding
- The Ethernet MAC Sublayer Protocol
- The Binary Exponential Backoff Algorithm
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet
- IEEE 802.2: Logical Link Control
- Retrospective on Ethernet

以太网电缆 (1)

Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings

The most common kinds of Ethernet cabling.

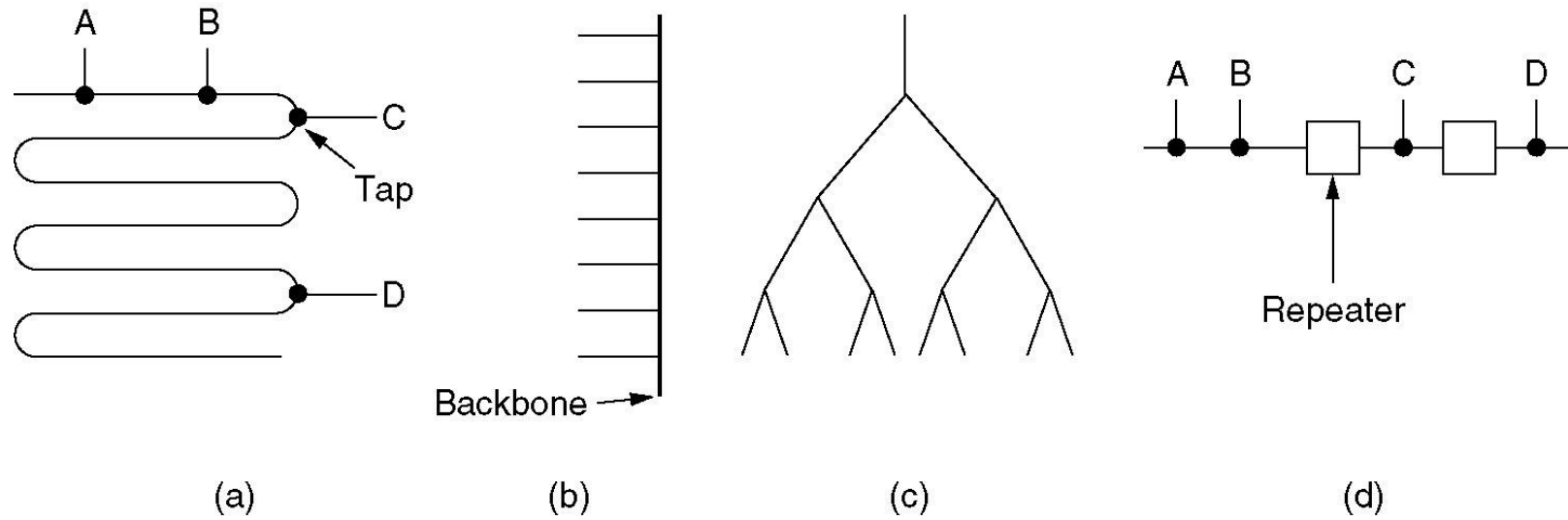
以太网电缆 (2)



Three kinds of Ethernet cabling.

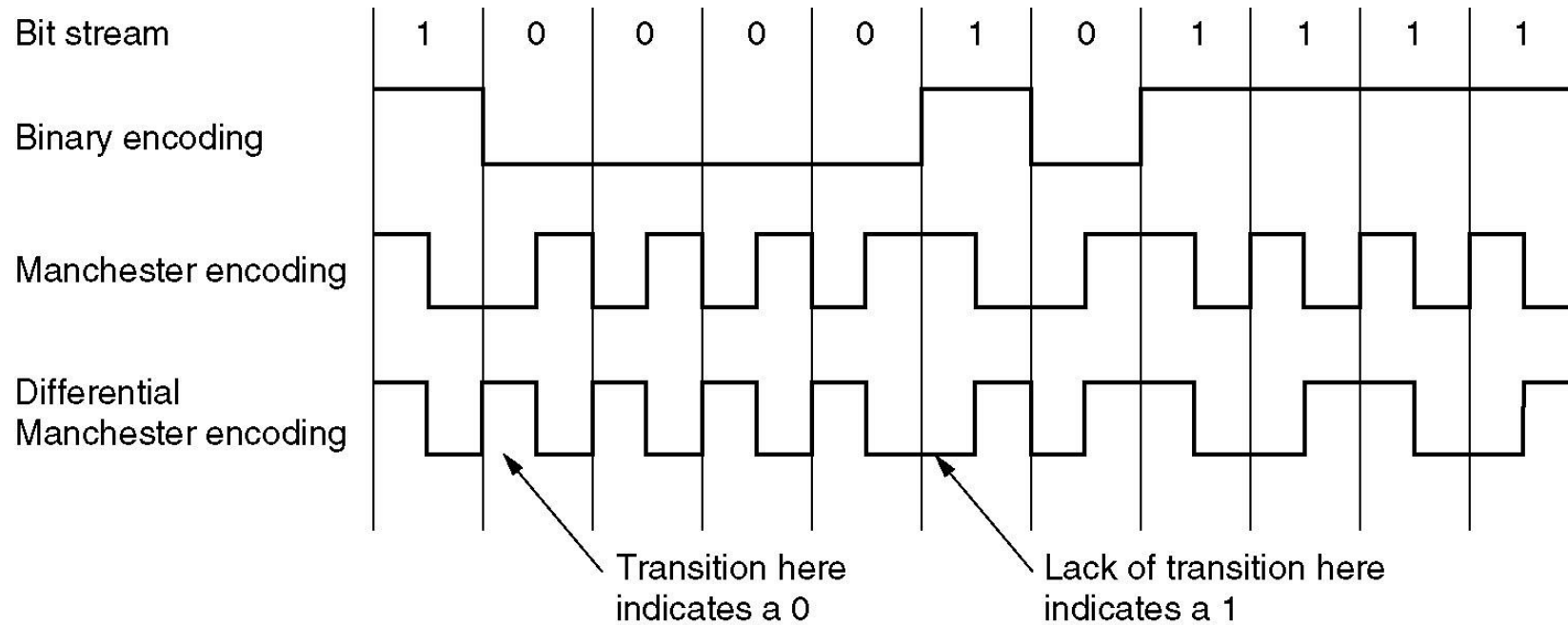
(a) 10Base5, (b) 10Base2, (c) 10Base-T.

以太网电缆 (3)



Cable topologies. (a) Linear, (b) Spine,
(c) Tree, (d) Segmented.

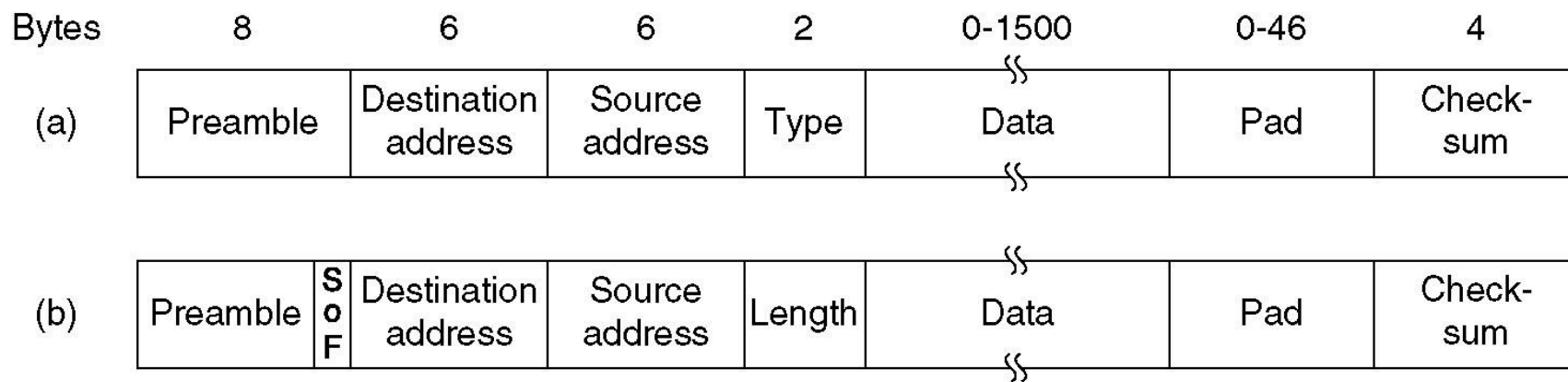
以太网电缆 (4)



(a) Binary encoding, (b) Manchester encoding,
(c) Differential Manchester encoding.

以太网介质访问子层协议 (1)

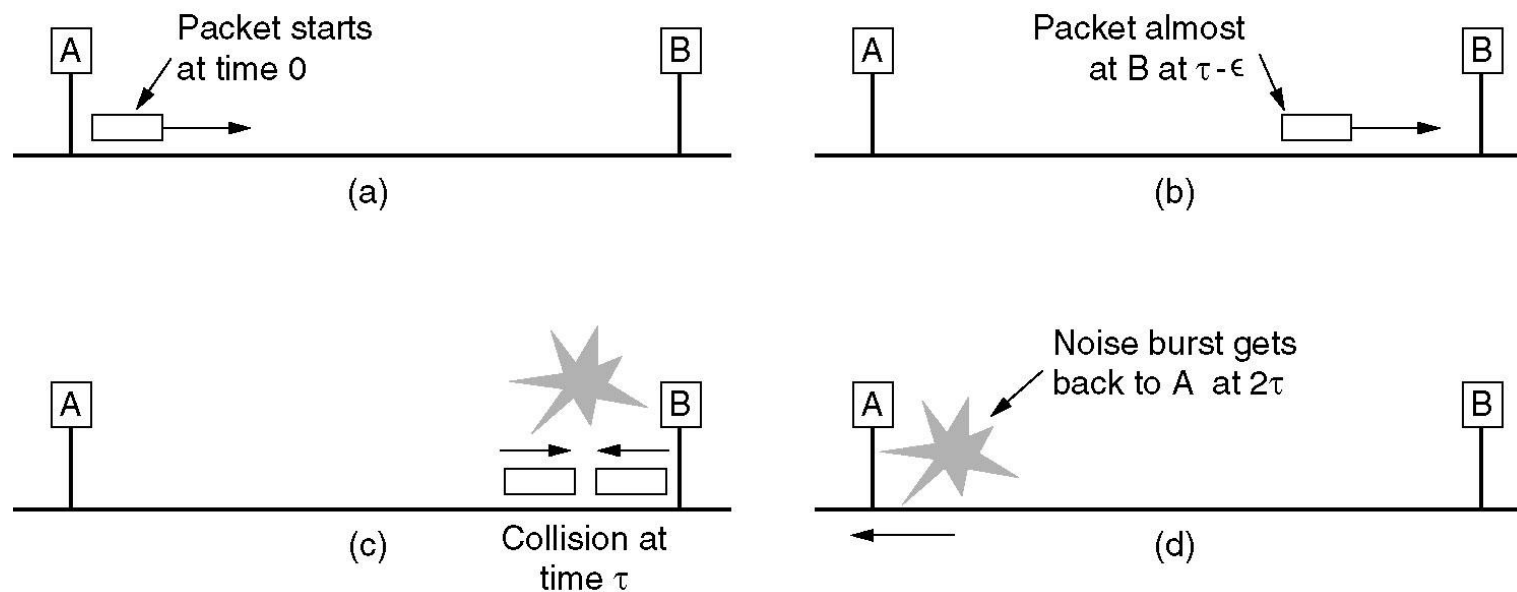
Ethernet MAC Sublayer Protocol



Frame formats. (a) DIX Ethernet, (b) IEEE 802.3.

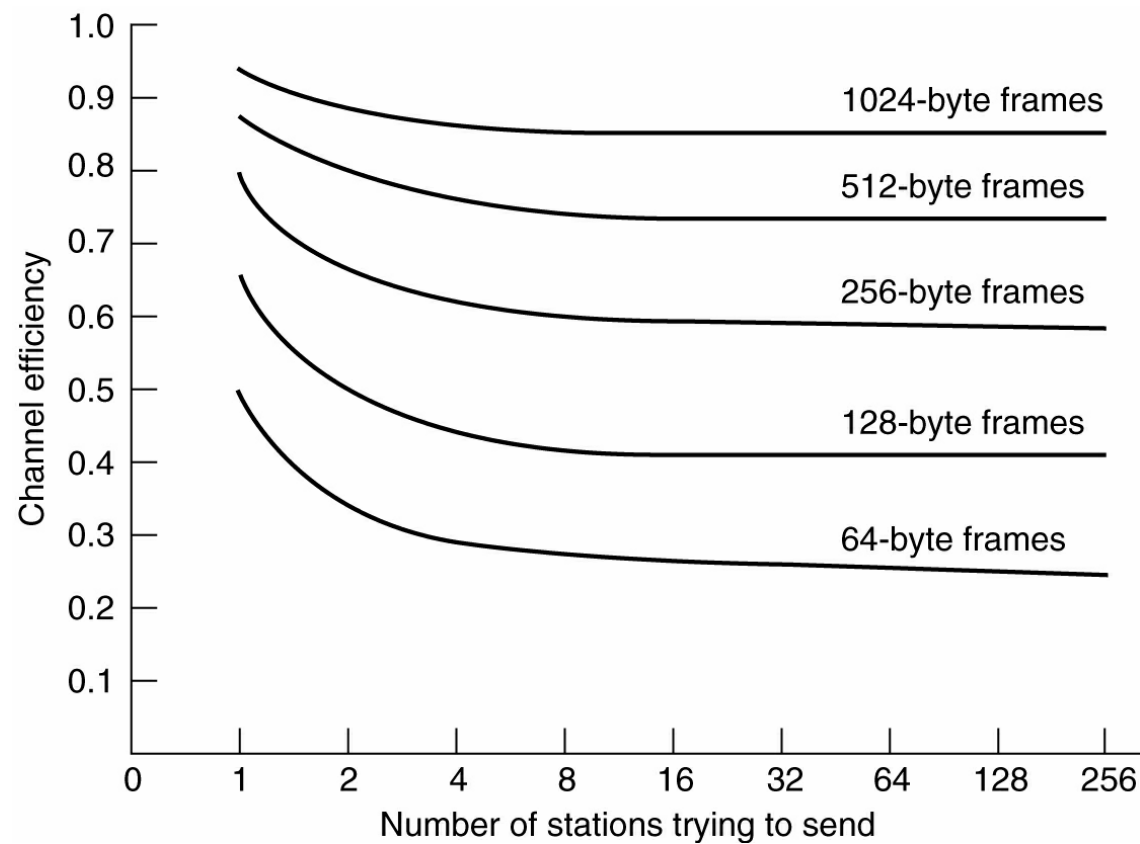
以太网介质访问子层协议 (2)

二进制退回到N重传协议



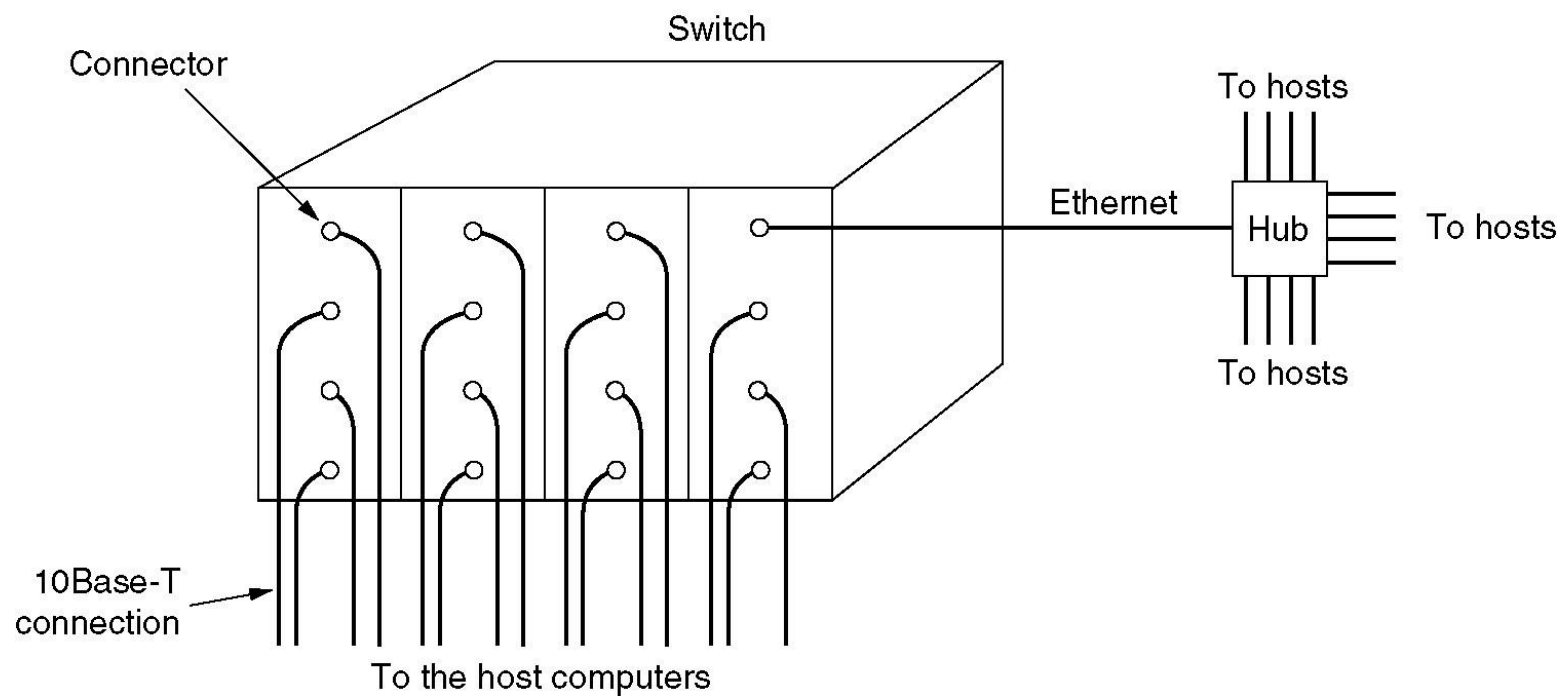
Collision detection can take as long as $2t$

以太网性能



Efficiency of Ethernet at 10 Mbps with 512-bit slot times.

交换以太网



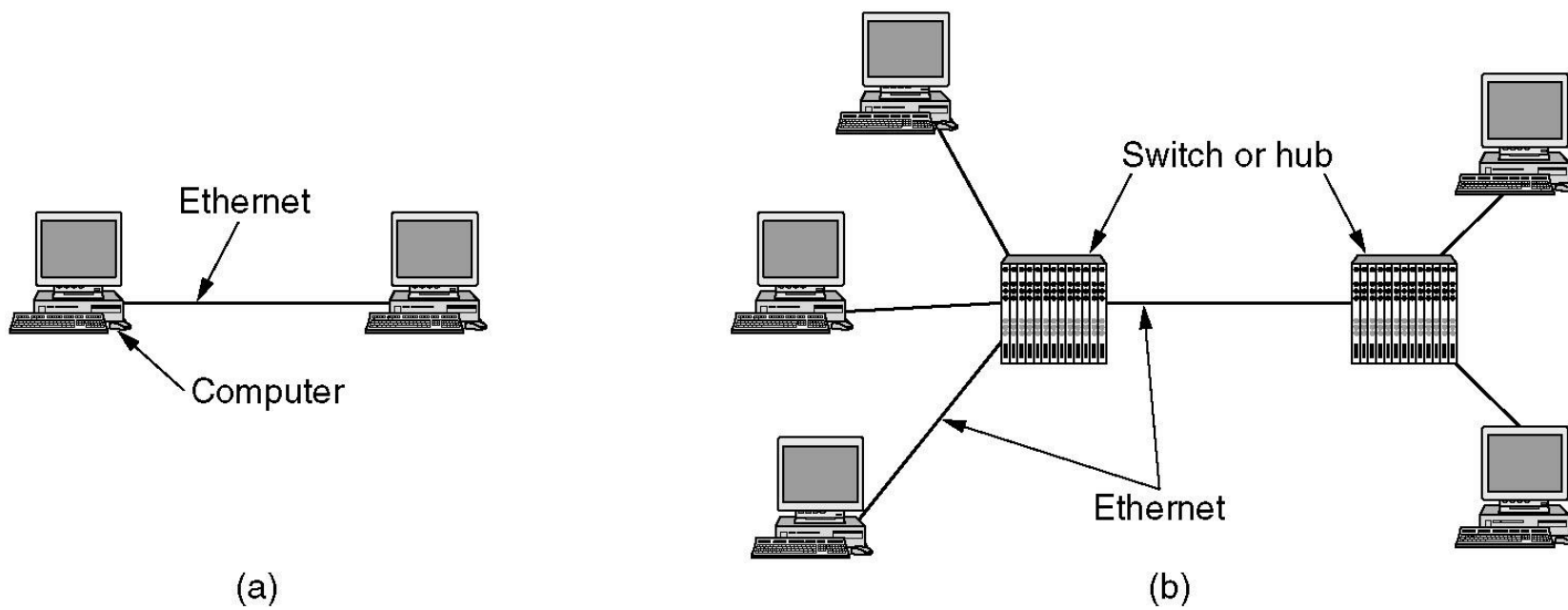
A simple example of switched Ethernet.

快速以太网

Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

The original fast Ethernet cabling.

千兆以太网 (1)



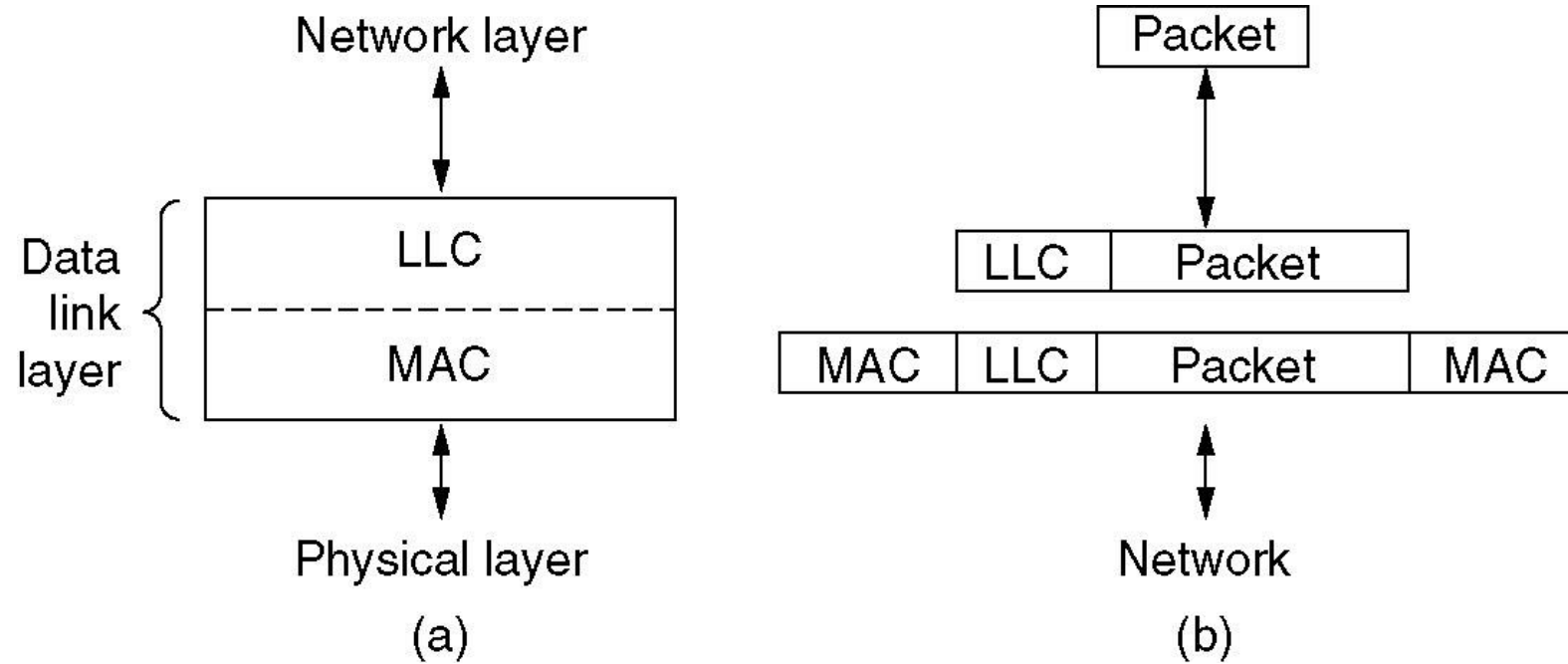
(a) A two-station Ethernet. (b) A multistation Ethernet.

千兆以太网 (2)

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

Gigabit Ethernet cabling.

IEEE 802.2: 逻辑链路控制

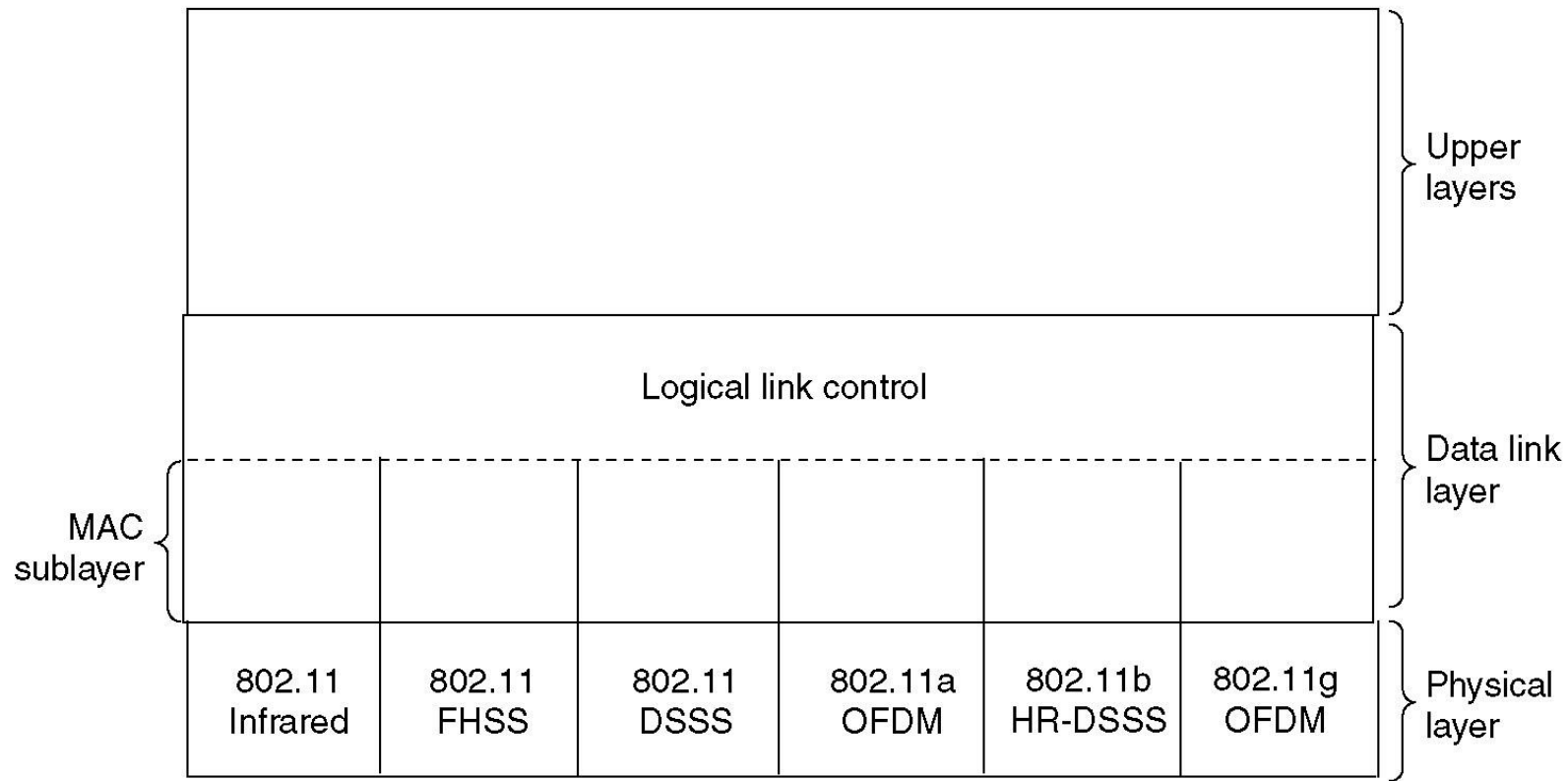


(a) Position of LLC. (b) Protocol formats.

6.4 无线局域网

- The 802.11 Protocol Stack
- The 802.11 Physical Layer
- The 802.11 MAC Sublayer Protocol
- The 802.11 Frame Structure
- Services

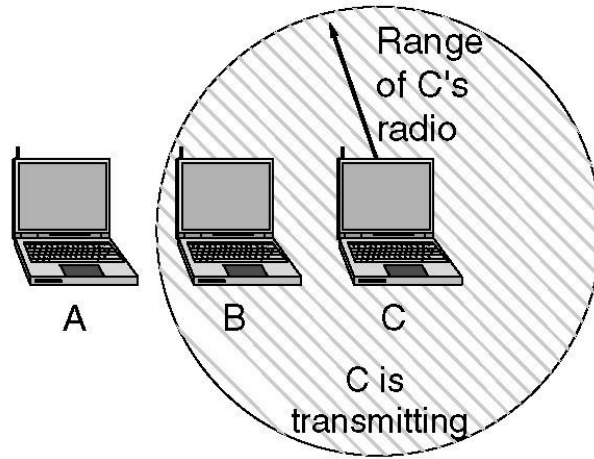
The 802.11 协议栈



Part of the 802.11 protocol stack.

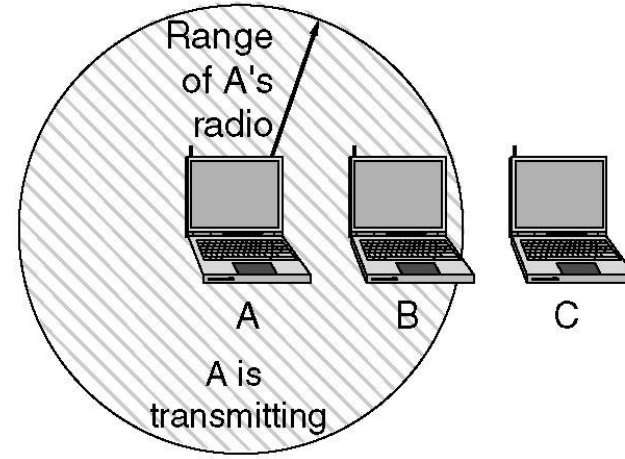
The 802.11 MAC 子层协议 (1)

A wants to send to B
but cannot hear that
B is busy



(a)

B wants to send to C
but mistakenly thinks
the transmission will fail

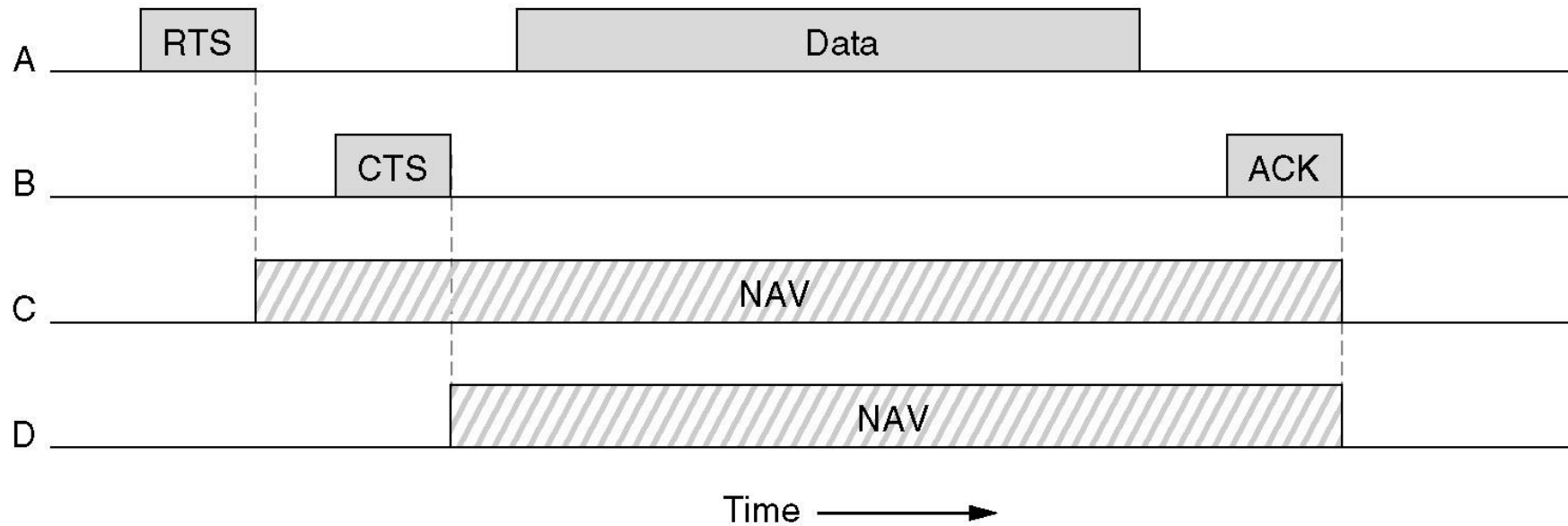


(b)

(a) The hidden station problem.

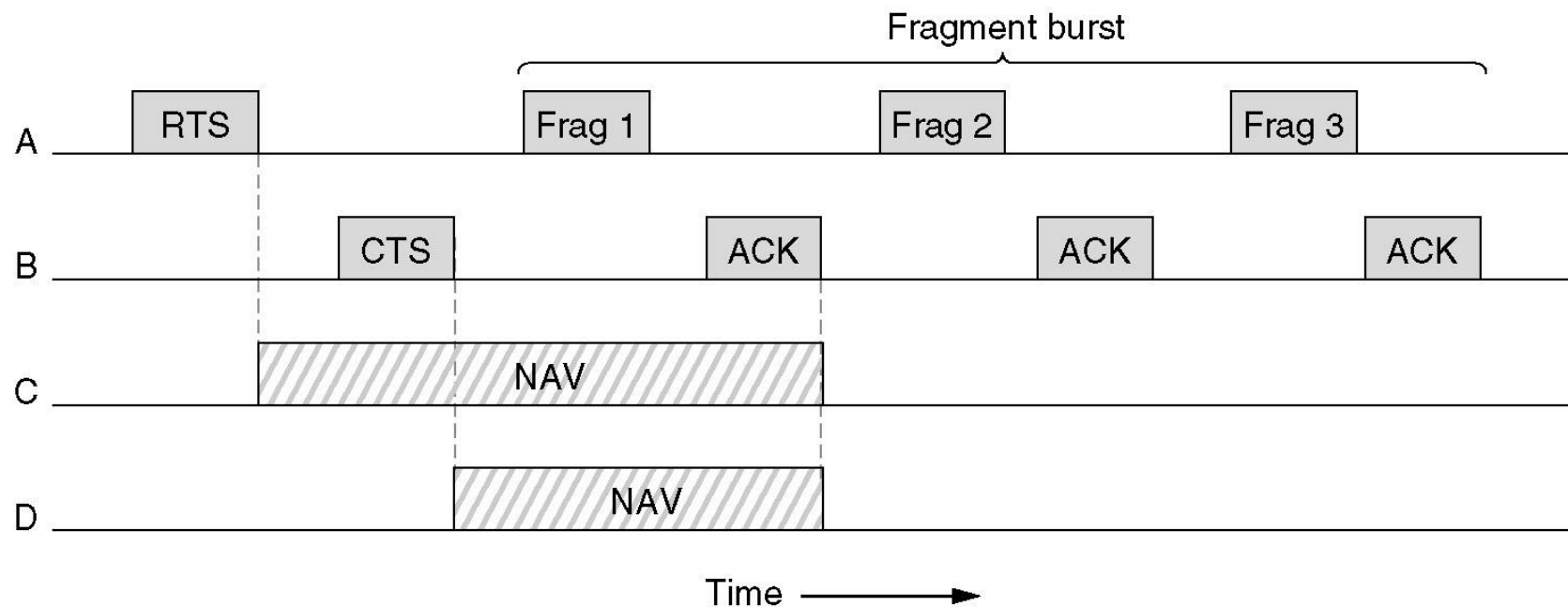
(b) The exposed station problem.

The 802.11 MAC 子层协议 (2)



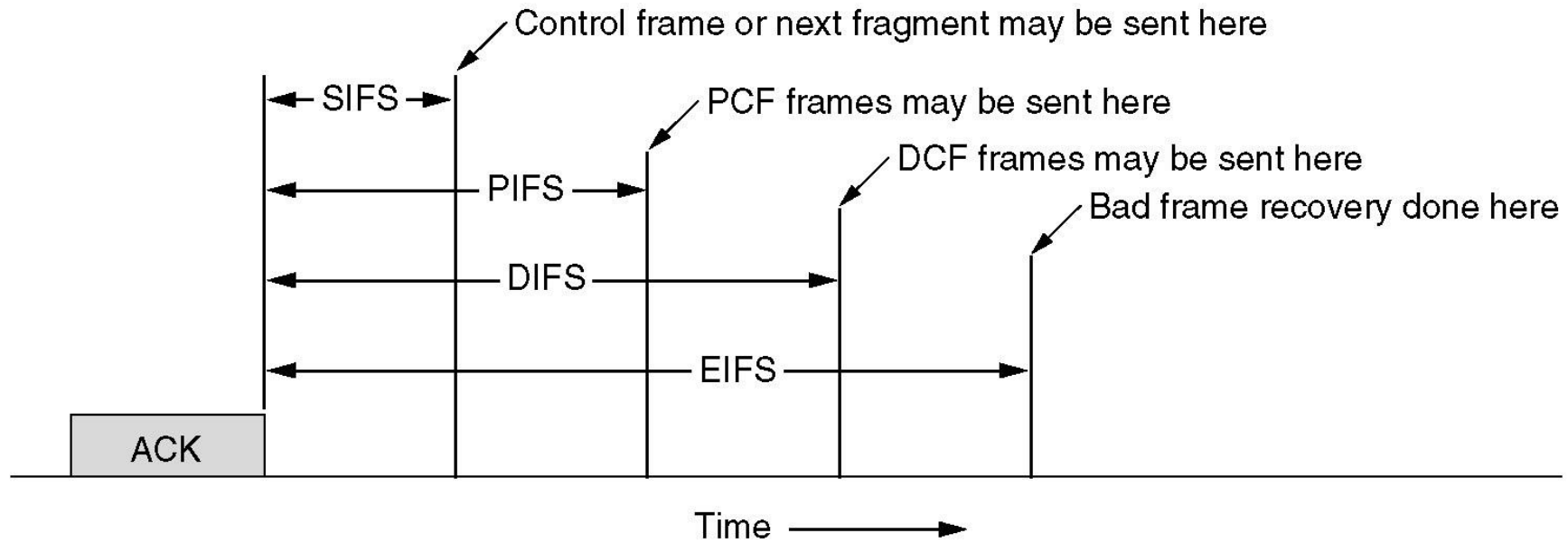
The use of virtual channel sensing using CSMA/CA.

The 802.11 MAC 子层协议 (3)



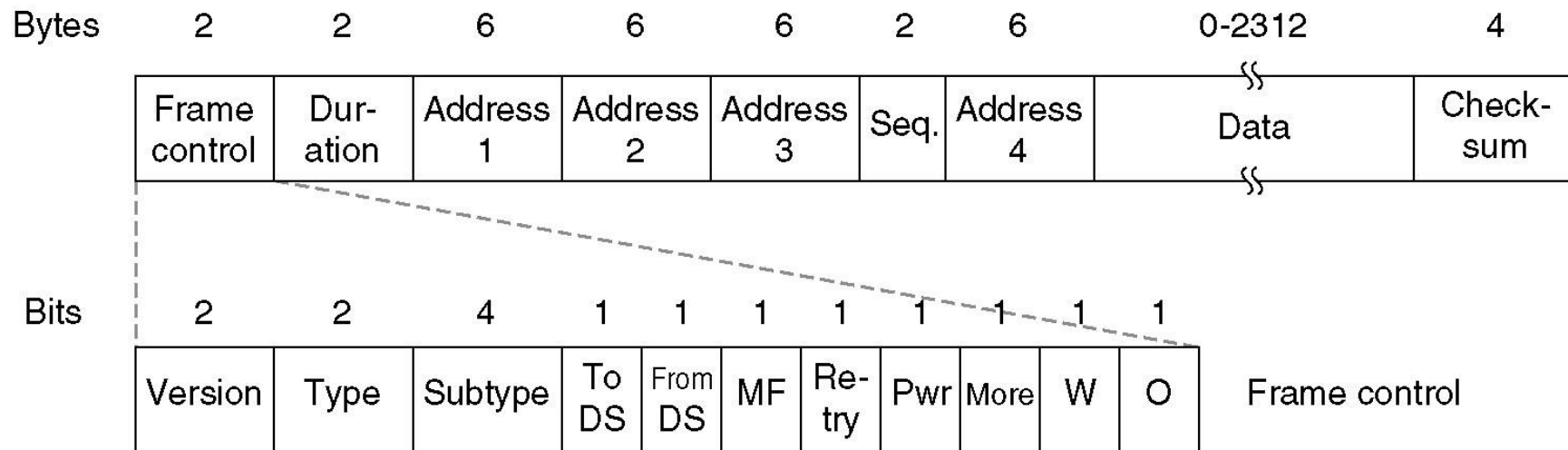
A fragment burst.

The 802.11 MAC 子层协议 (4)



Interframe spacing in 802.11.

The 802.11 MAC 帧结构



The 802.11 data frame.

802.11 服务

Distribution Services

- Association
- Disassociation
- Reassociation
- Distribution
- Integration

802.11 服务

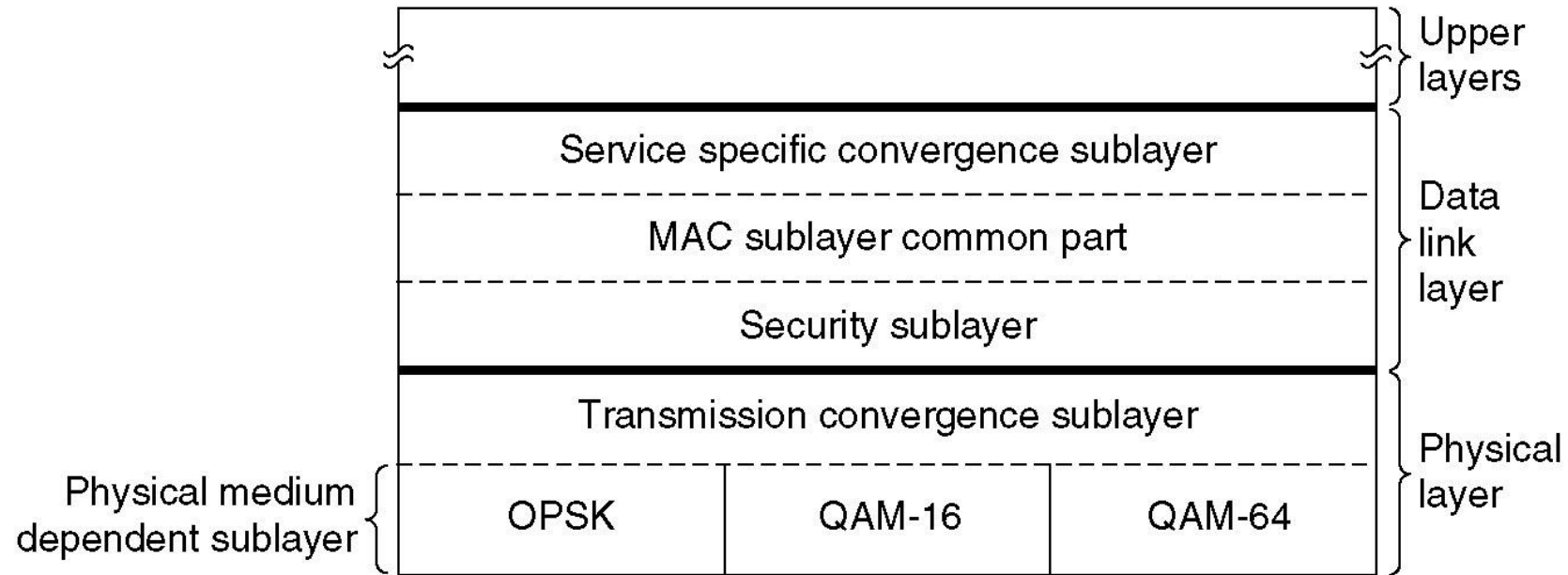
Intracell Services

- Authentication
- Deauthentication
- Privacy
- Data Delivery

6.5 宽带无线局域网

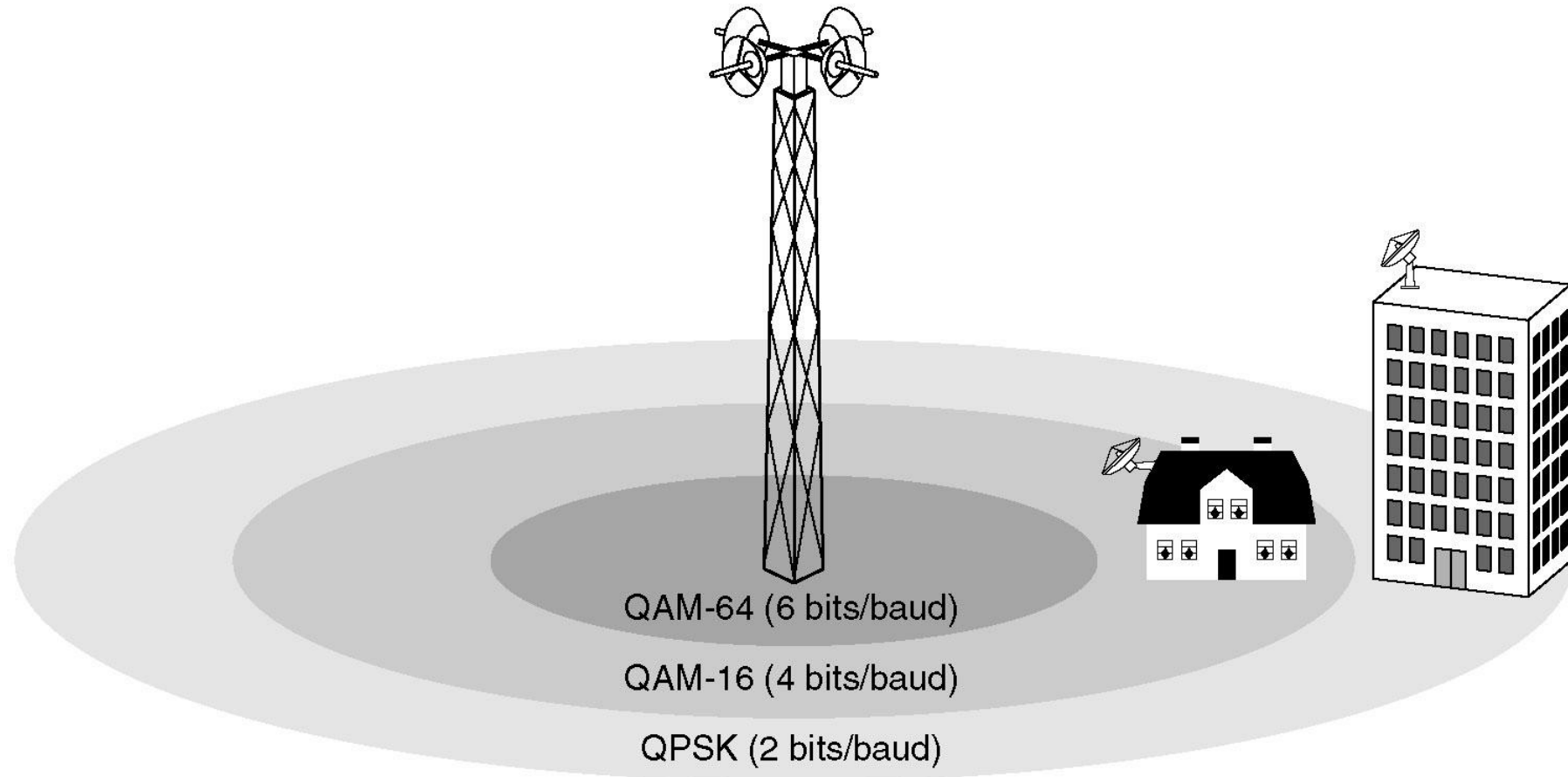
- Comparison of 802.11 and 802.16
- The 802.16 Protocol Stack
- The 802.16 Physical Layer
- The 802.16 MAC Sublayer Protocol
- The 802.16 Frame Structure

The 802.16 Protocol Stack



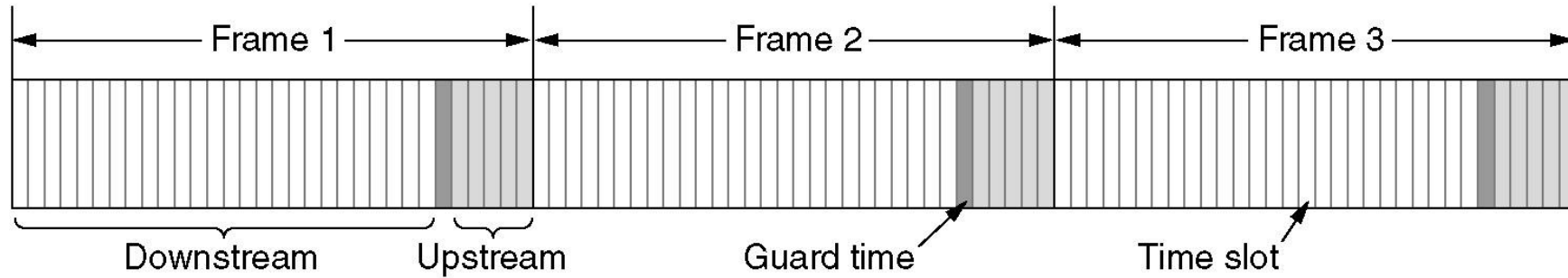
The 802.16 Protocol Stack.

The 802.16 Physical Layer



The 802.16 transmission environment.

The 802.16 Physical Layer (2)



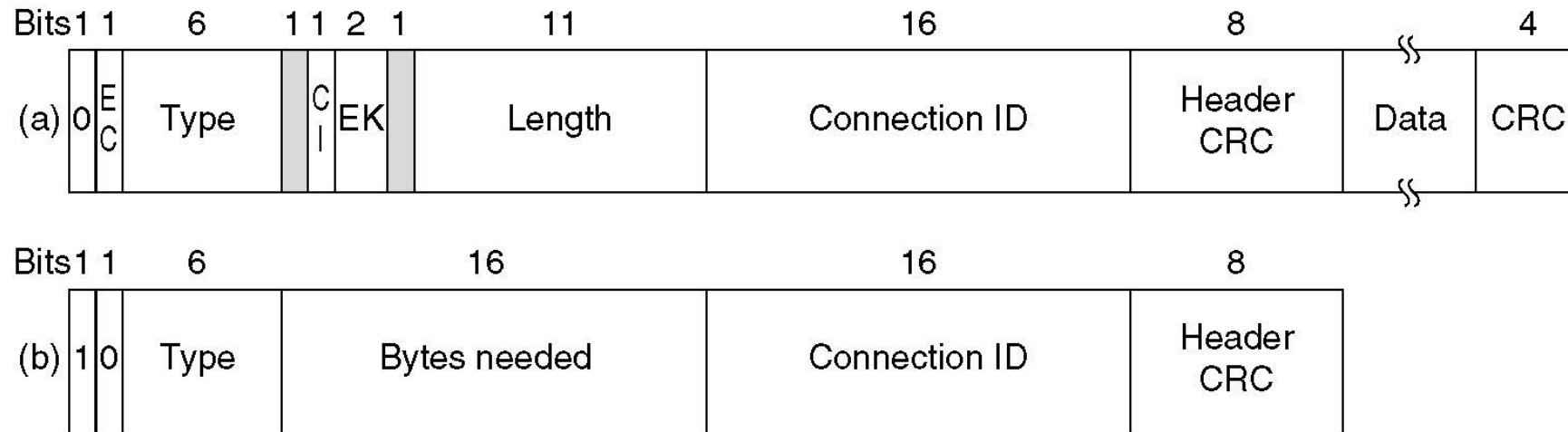
Frames and time slots for time division duplexing.

The 802.16 MAC Sublayer Protocol

Service Classes

- Constant bit rate service
- Real-time variable bit rate service
- Non-real-time variable bit rate service
- Best efforts service

The 802.16 Frame Structure

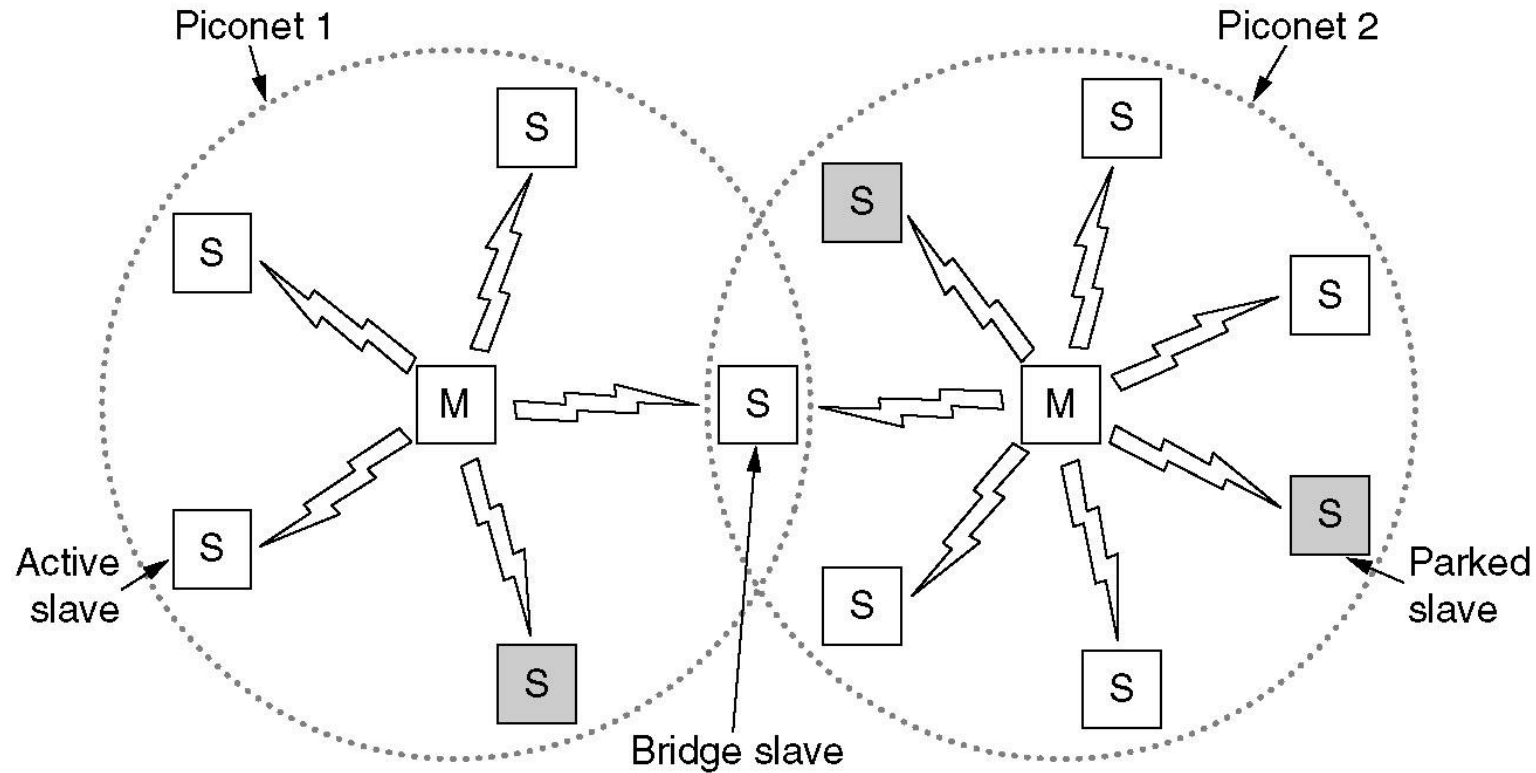


(a) A generic frame. (b) A bandwidth request frame.

Bluetooth

- Bluetooth Architecture
- Bluetooth Applications
- The Bluetooth Protocol Stack
- The Bluetooth Radio Layer
- The Bluetooth Baseband Layer
- The Bluetooth L2CAP Layer
- The Bluetooth Frame Structure

Bluetooth Architecture



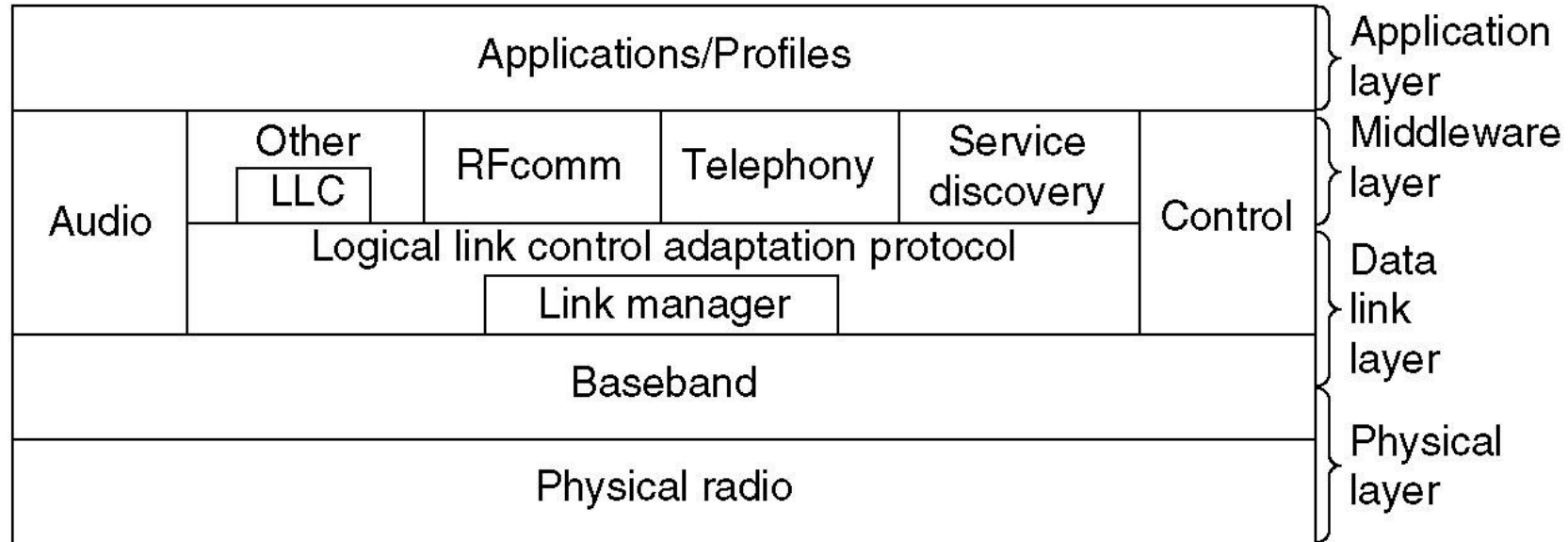
Two piconets can be connected to form a scatternet.

Bluetooth Applications

Name	Description
Generic access	Procedures for link management
Service discovery	Protocol for discovering offered services
Serial port	Replacement for a serial port cable
Generic object exchange	Defines client-server relationship for object movement
LAN access	Protocol between a mobile computer and a fixed LAN
Dial-up networking	Allows a notebook computer to call via a mobile phone
Fax	Allows a mobile fax machine to talk to a mobile phone
Cordless telephony	Connects a handset and its local base station
Intercom	Digital walkie-talkie
Headset	Intended for hands-free voice communication
Object push	Provides a way to exchange simple objects
File transfer	Provides a more general file transfer facility
Synchronization	Permits a PDA to synchronize with another computer

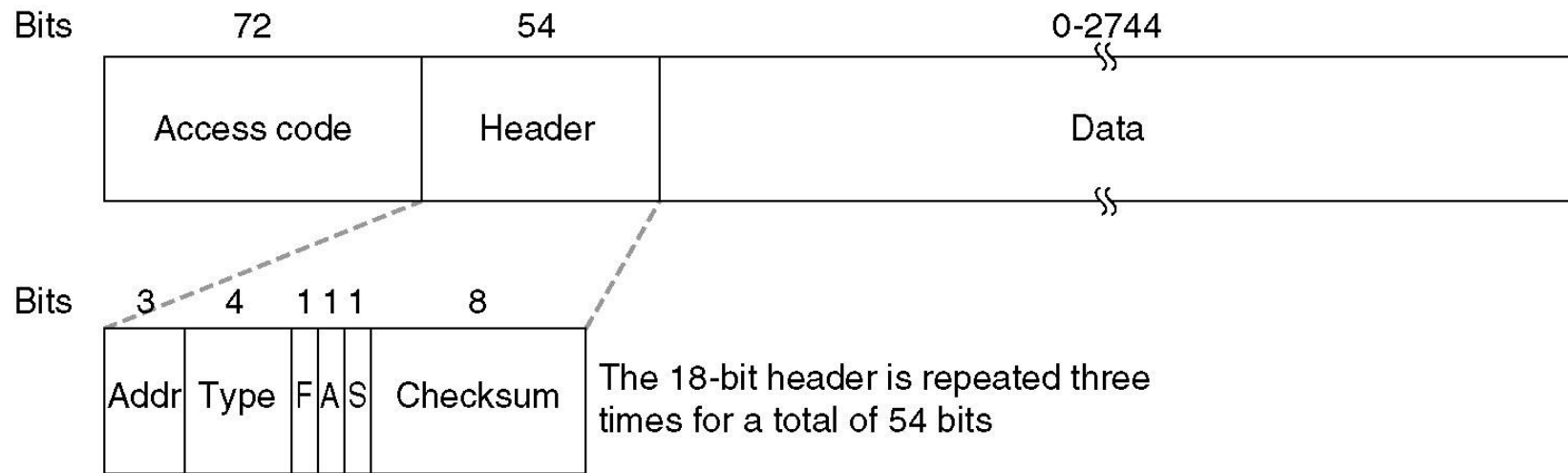
The Bluetooth profiles.

The Bluetooth Protocol Stack



The 802.15 version of the Bluetooth protocol architecture.

The Bluetooth Frame Structure

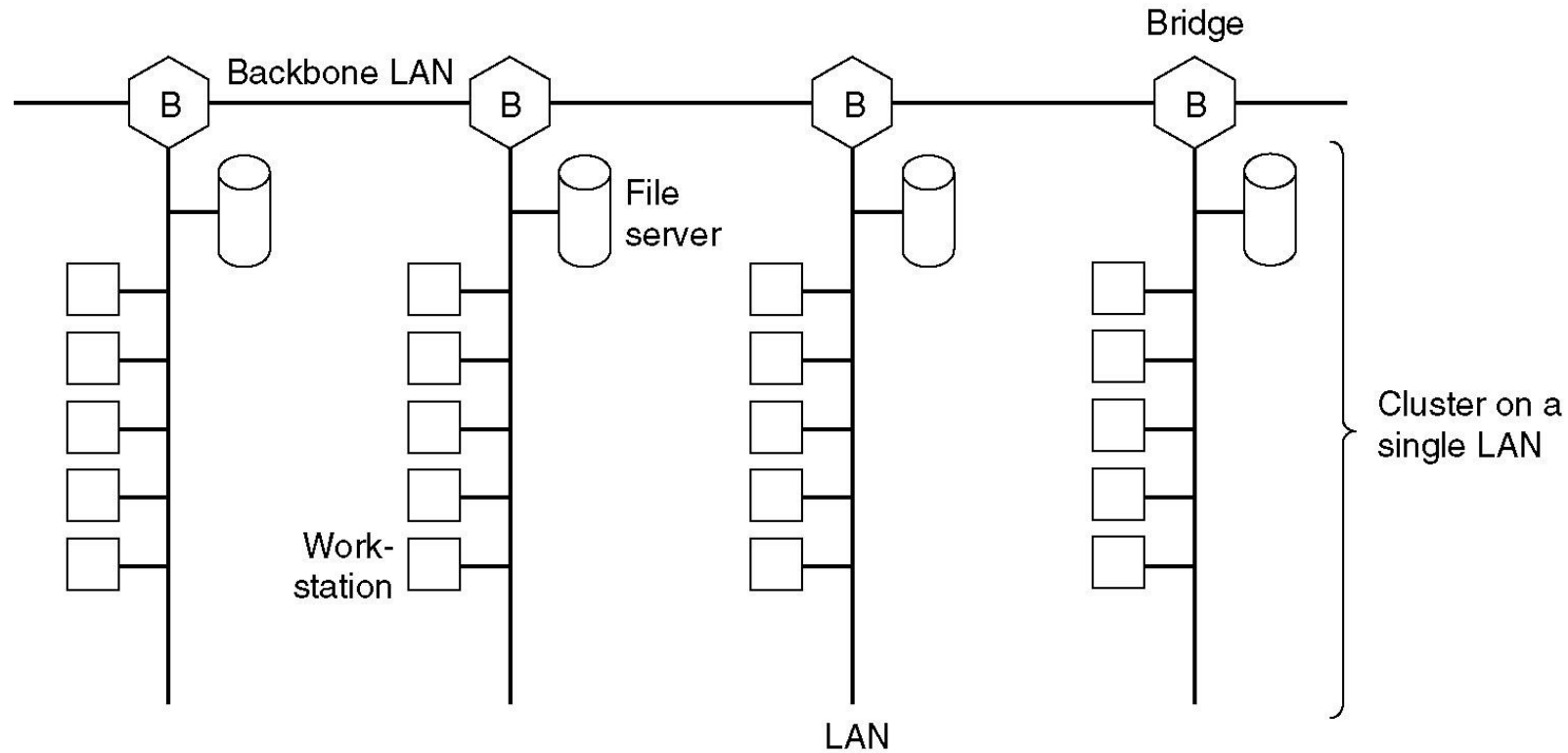


A typical Bluetooth data frame.

6.6 Data Link Layer Switching

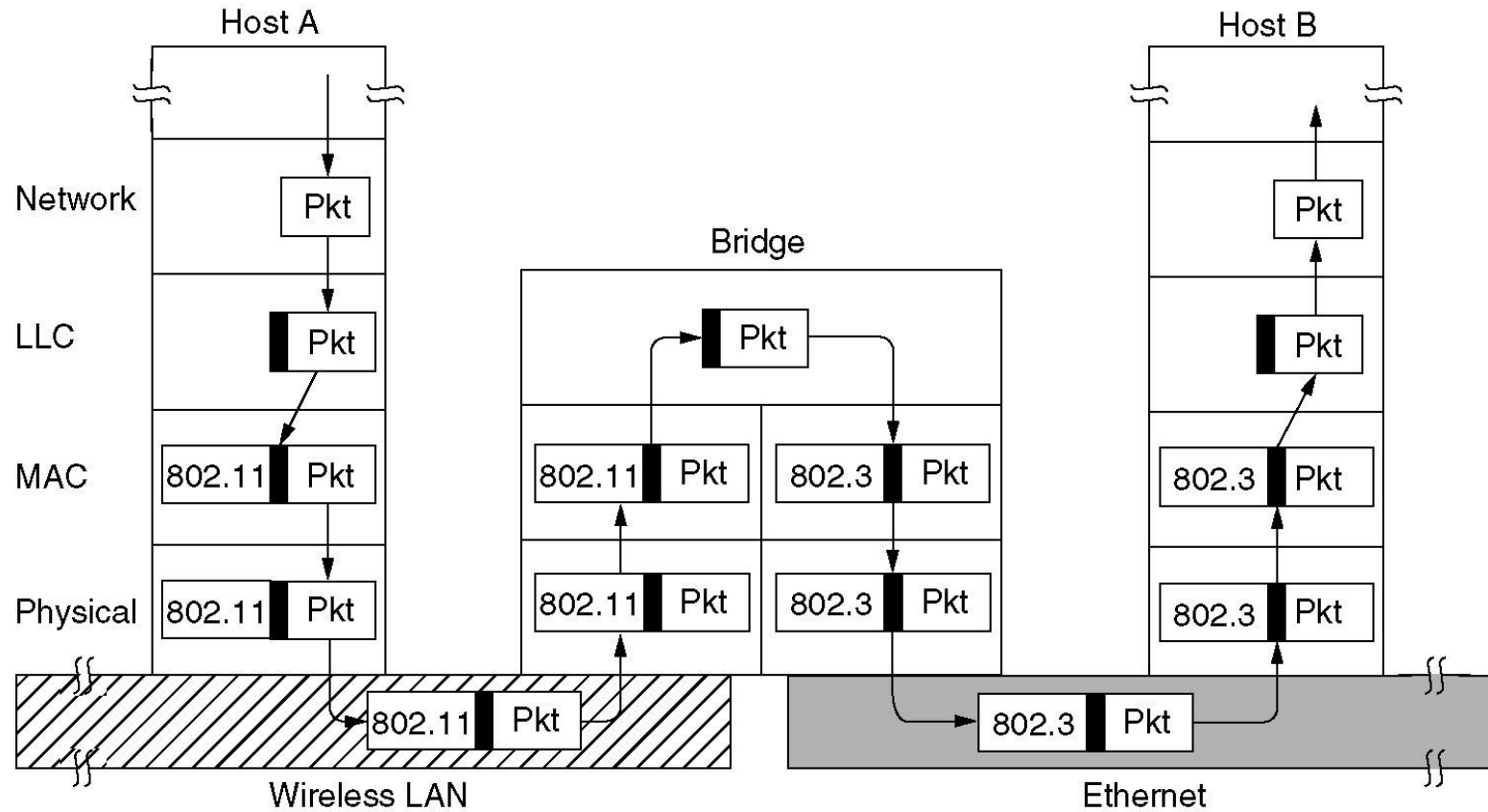
- Bridges from 802.x to 802.y
- Local Internetworking
- Spanning Tree Bridges
- Remote Bridges
- Repeaters, Hubs, Bridges, Switches, Routers, Gateways
- Virtual LANs

Data Link Layer Switching



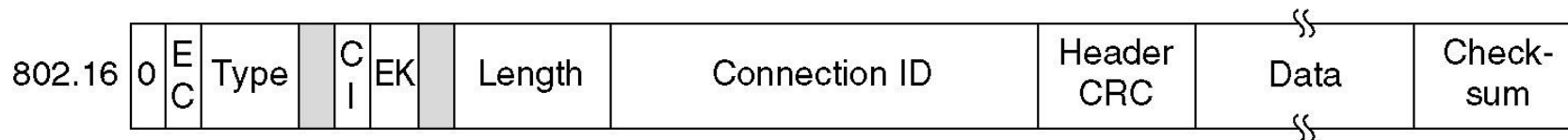
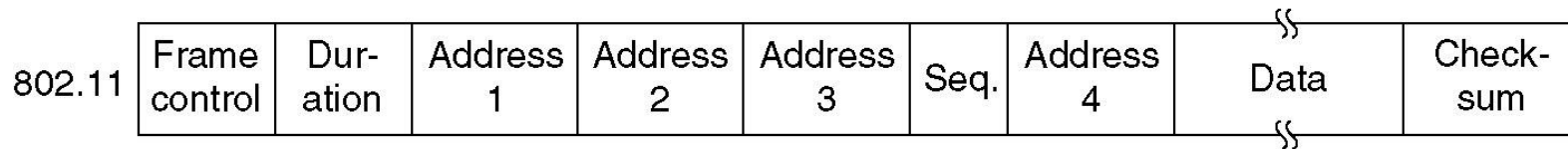
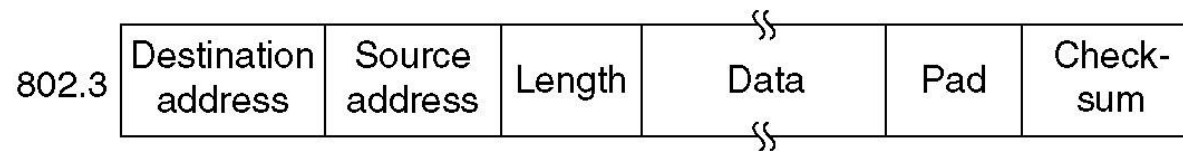
Multiple LANs connected by a backbone to handle a total load higher than the capacity of a single LAN.

Bridges from 802.x to 802.y



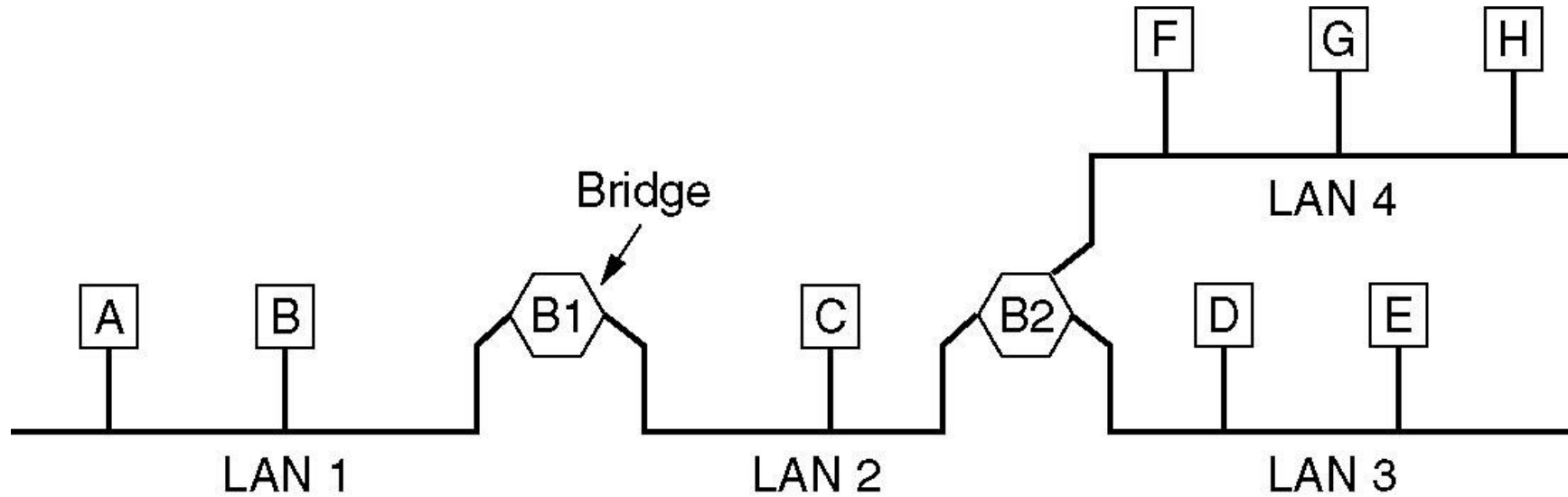
Operation of a LAN bridge from 802.11 to 802.3.

Bridges from 802.x to 802.y (2)



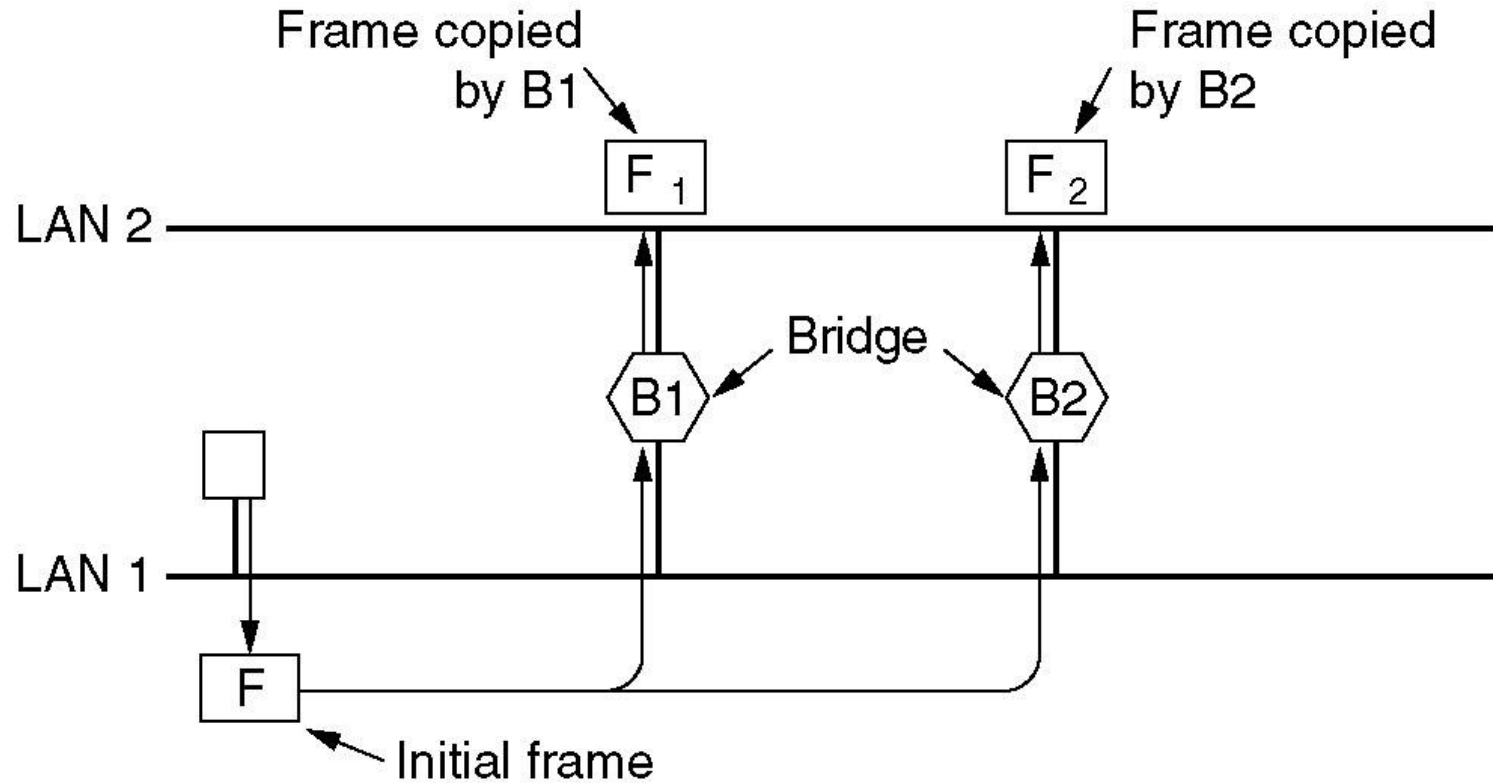
The IEEE 802 frame formats. The drawing is not to scale.

Local Internetworking



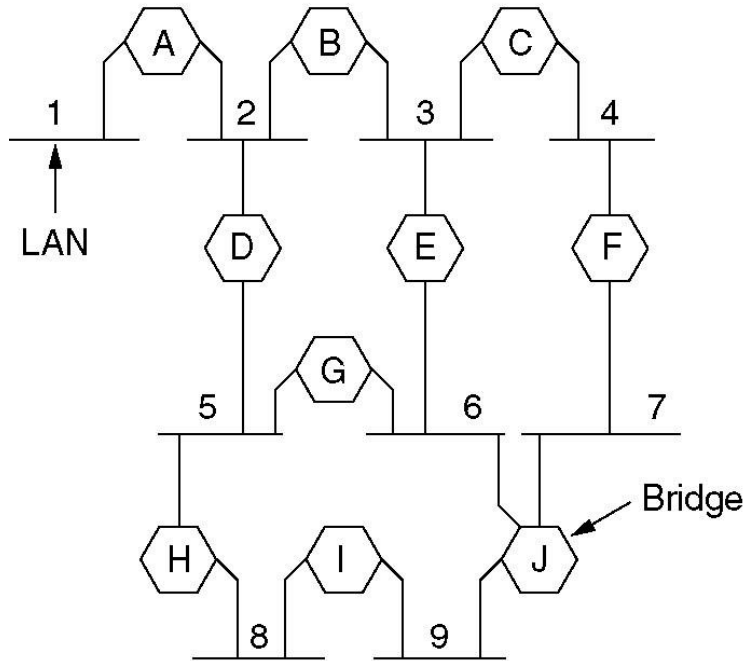
A configuration with four LANs and two bridges.

Spanning Tree Bridges

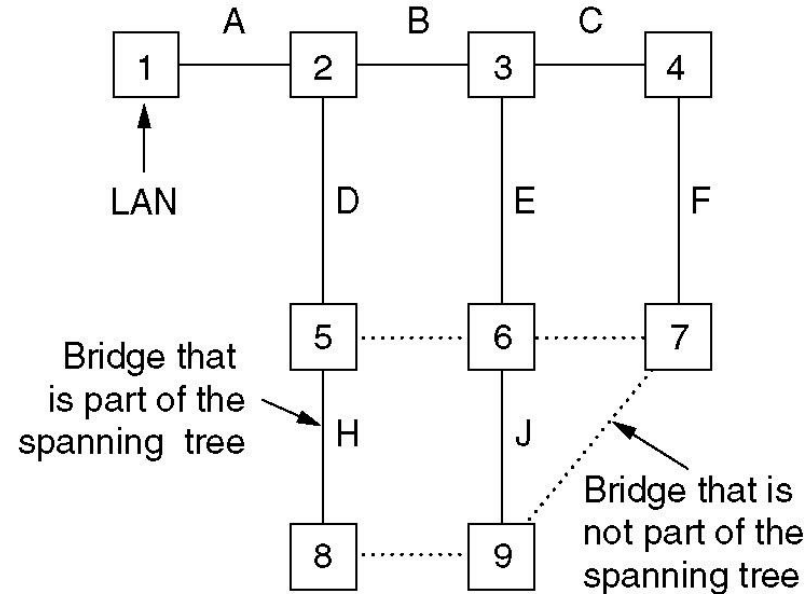


Two parallel transparent bridges.

Spanning Tree Bridges (2)



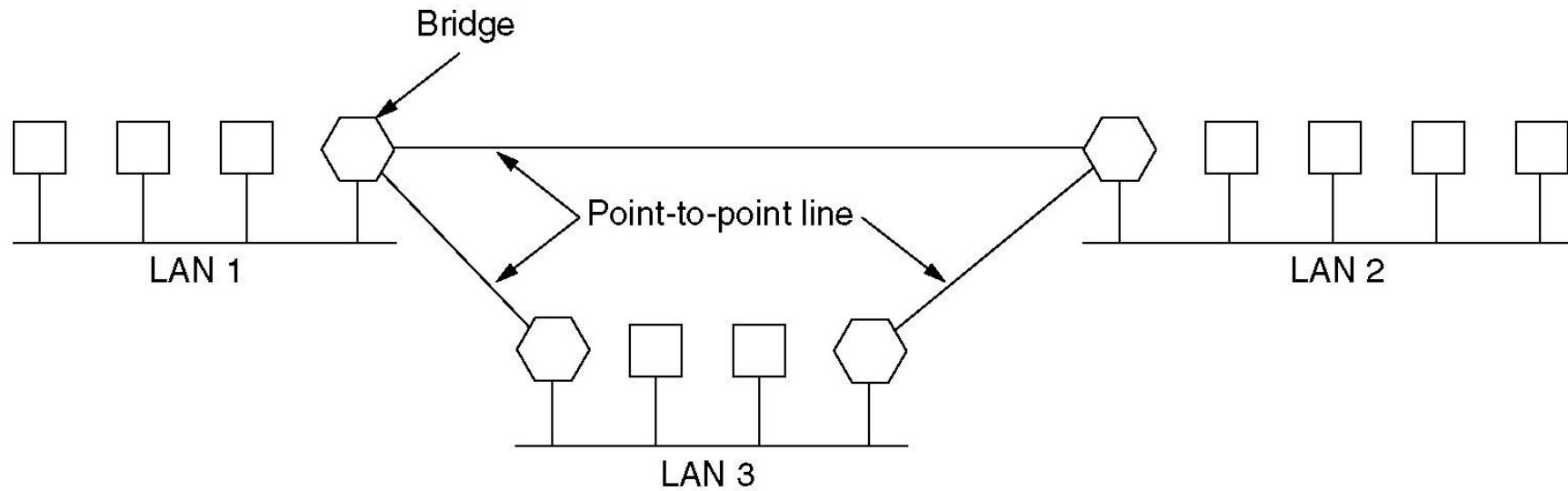
(a)



(b)

(a) Interconnected LANs. (b) A spanning tree covering the LANs. The dotted lines are not part of the spanning tree.

Remote Bridges

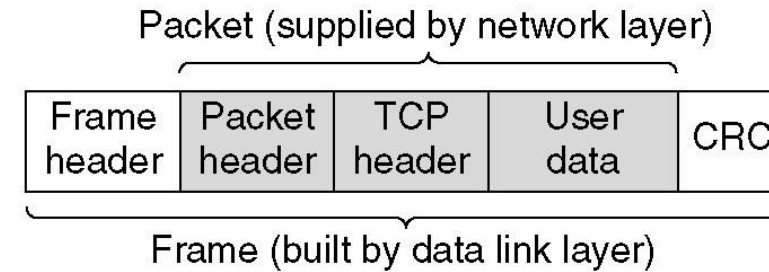


Remote bridges can be used to interconnect distant LANs.

Repeaters, Hubs, Bridges, Switches, Routers and Gateways

Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

(a)

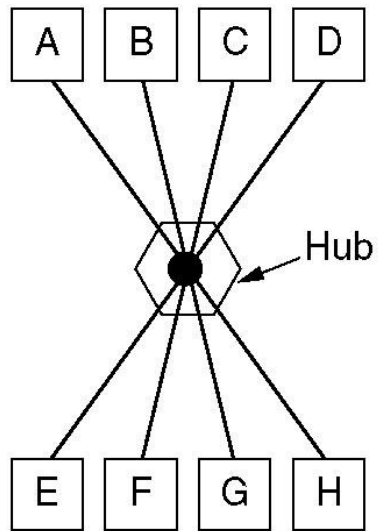


(b)

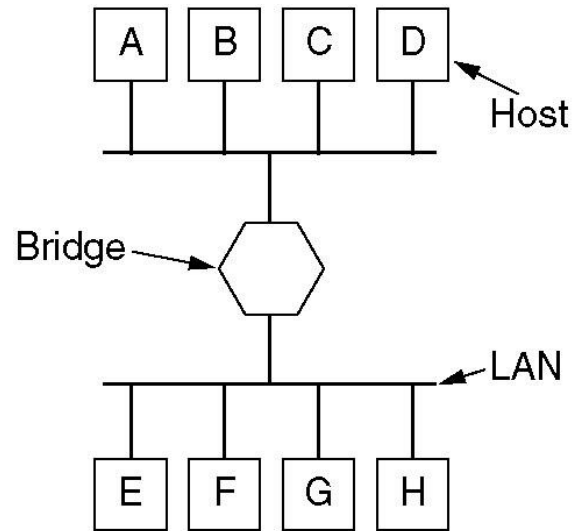
(a) Which device is in which layer.

(b) Frames, packets, and headers.

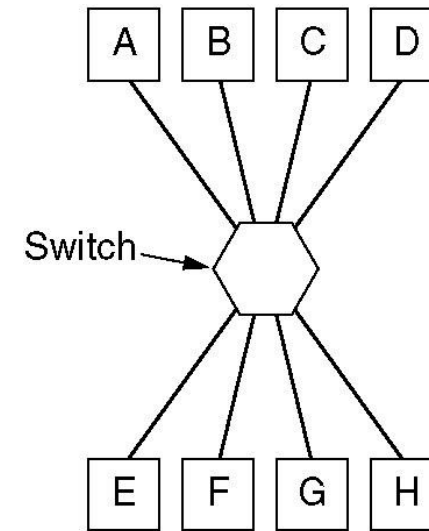
Repeaters, Hubs, Bridges, Switches, Routers and Gateways (2)



(a)



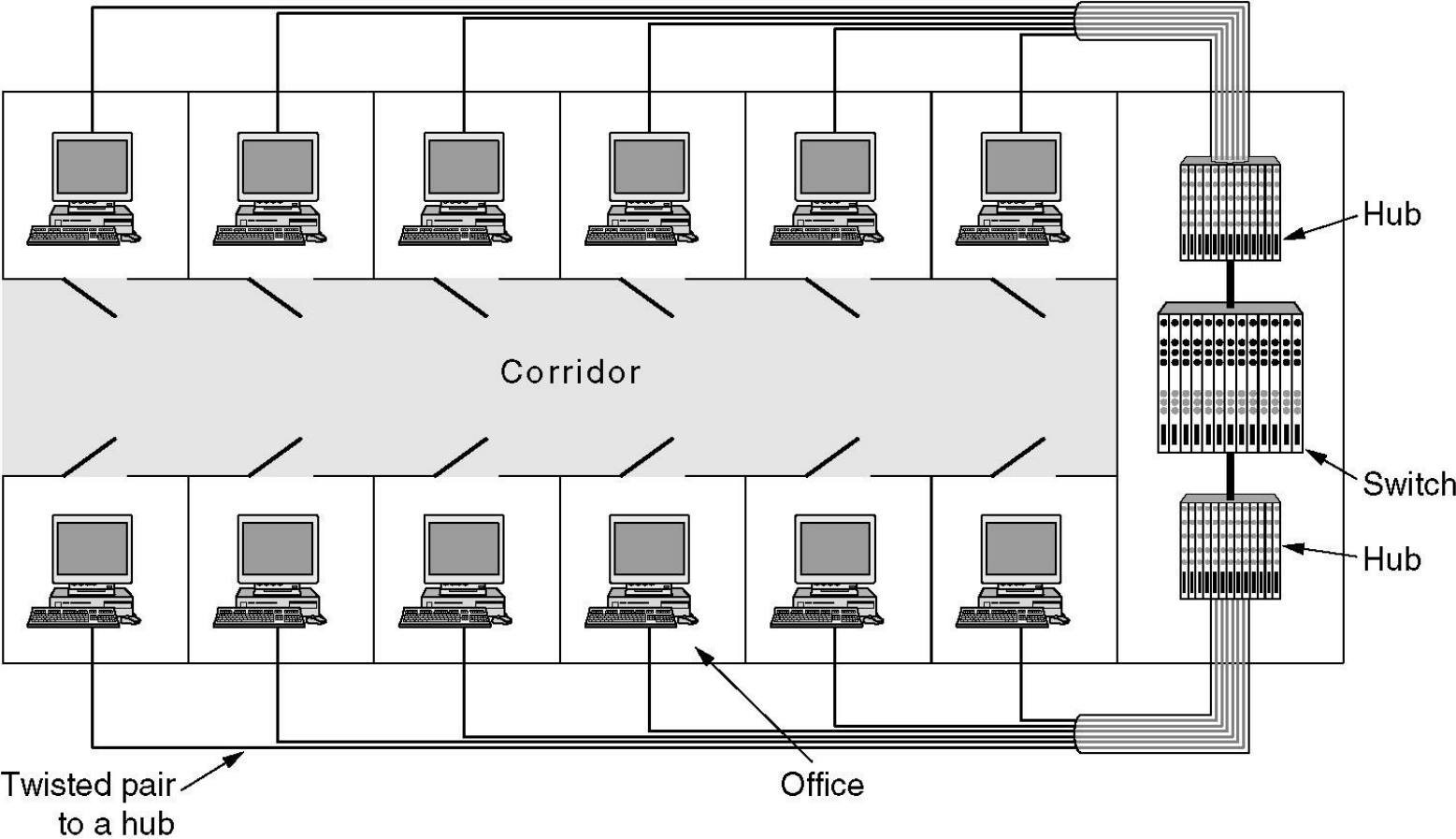
(b)



(c)

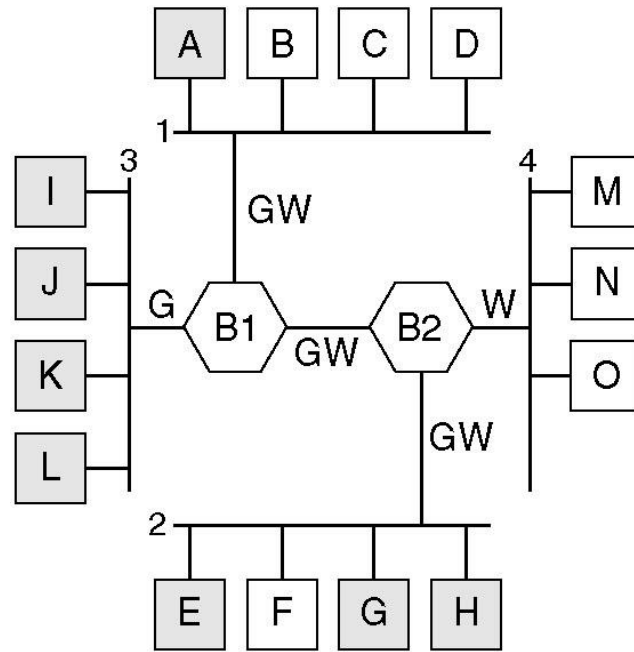
(a) A hub. (b) A bridge. (c) a switch.

Virtual LANs

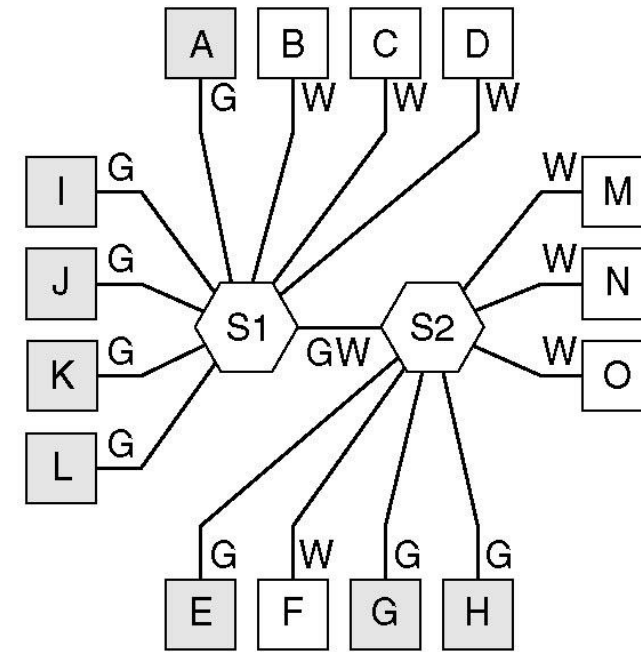


A building with centralized wiring using hubs and a switch.

Virtual LANs (2)



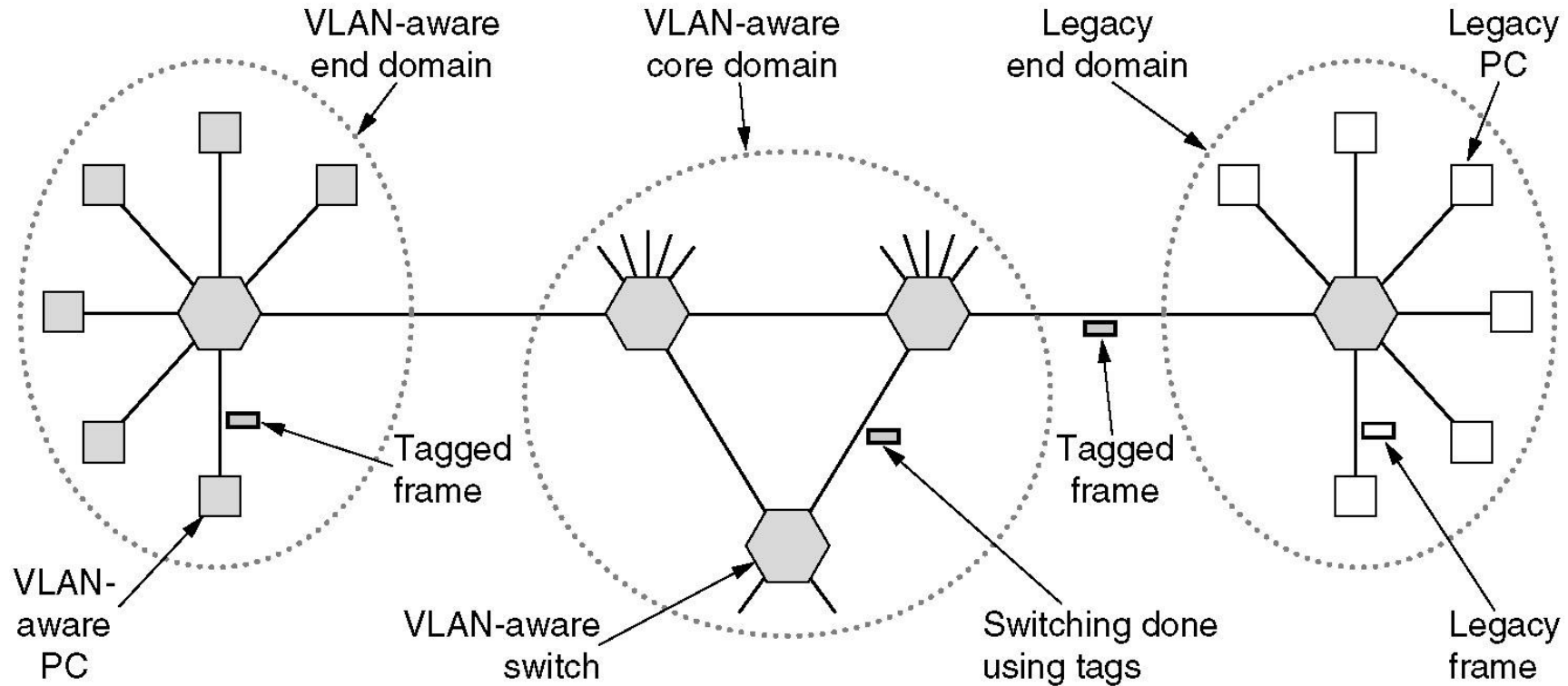
(a)



(b)

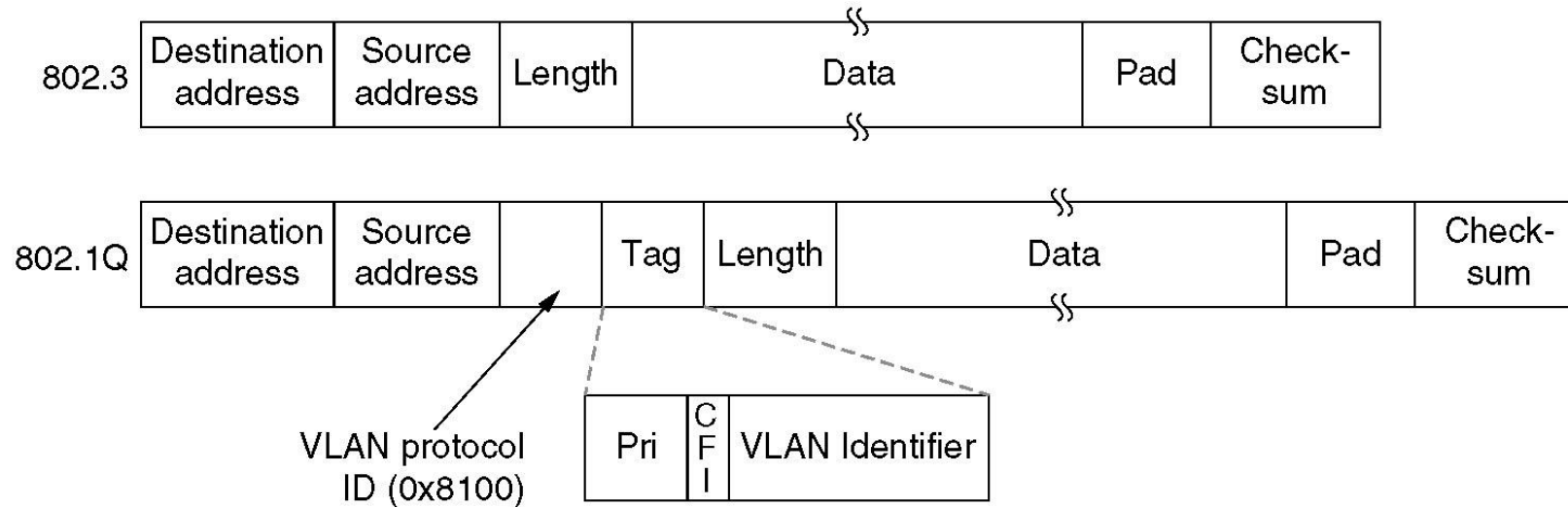
(a) Four physical LANs organized into two VLANs, gray and white, by two bridges. (b) The same 15 machines organized into two VLANs by switches.

The IEEE 802.1Q Standard



Transition from legacy Ethernet to VLAN-aware Ethernet.
The shaded symbols are VLAN aware. The empty ones are not.

The IEEE 802.1Q Standard (2)



The 802.3 (legacy) and 802.1Q Ethernet frame formats.

本章小结

Method	Description
FDM	Dedicate a frequency band to each station
WDM	A dynamic FDM scheme for fiber
TDM	Dedicate a time slot to each station
Pure ALOHA	Unsynchronized transmission at any instant
Slotted ALOHA	Random transmission in well-defined time slots
1-persistent CSMA	Standard carrier sense multiple access
Nonpersistent CSMA	Random delay when channel is sensed busy
P-persistent CSMA	CSMA, but with a probability of p of persisting
CSMA/CD	CSMA, but abort on detecting a collision
Bit map	Round robin scheduling using a bit map
Binary countdown	Highest numbered ready station goes next
Tree walk	Reduced contention by selective enabling
MACA, MACAW	Wireless LAN protocols
Ethernet	CSMA/CD with binary exponential backoff
FHSS	Frequency hopping spread spectrum
DSSS	Direct sequence spread spectrum
CSMA/CA	Carrier sense multiple access with collision avoidance

Channel allocation methods and systems for a common channel.

本章结束